

# HSC Math Extension 2: Complex Numbers Mastery

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*Download PDF and resources:*

<https://github.com/vuhung16au/math-olympiad-ml>

*“Pure mathematics is, in its way, the poetry of logical ideas.”*

Albert Einstein

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# 1 Introduction

## 1.1 Project Overview

This booklet compiles high quality complex numbers problems curated specifically for the HSC Mathematics Extension 2 syllabus. Every problem explores fundamental and advanced techniques involving complex numbers—from basic arithmetic and form conversions to De Moivre’s theorem, Argand diagram geometry, polynomial roots, and geometric transformations. Detailed reasoning showcases common techniques such as modulus-argument manipulation, Euler’s formula applications, and geometric interpretations.

## 1.2 Target Audience

The explanations are crafted for Extension 2 students aiming to deepen their complex number skills. Each solution explicitly states the conversion steps, theorem applications, and geometric reasoning so that high-school learners can follow every transition.

## 1.3 How to Use This Booklet

- Read the overview and complex numbers primer before attempting the problems.
- Attempt problems in Part 1 without hints; compare against the detailed solutions to understand model reasoning.
- Use the upside-down hints in Part 2 only after making a genuine attempt.
- Revisit problems after a few days and try to re-derive the arguments without notes to reinforce technique.

## 1.4 Key Topics Covered

This booklet comprehensively covers the following essential topics from the HSC Mathematics Extension 2 Complex Numbers syllabus:

- **Arithmetic and Quadratic Equations:** Operations with complex numbers, solving quadratic equations with complex roots, and understanding the relationship between coefficients and roots.
- **Argand Diagram:** Visual representation of complex numbers as points or vectors in the complex plane, interpreting geometric relationships and transformations.
- **Euler’s Theorem:** The fundamental relationship  $e^{i\theta} = \cos \theta + i \sin \theta$  connecting exponential, trigonometric, and complex number representations. *Note:* A major syllabus change is that Euler’s form ( $re^{i\theta}$ ) has been completely removed from the syllabus; it is not examined, but it remains useful and closely connected to higher-education mathematics—keep it in mind.
- **De Moivre’s Theorem:** Computing powers and roots of complex numbers using  $(r(\cos \theta + i \sin \theta))^n = r^n(\cos(n\theta) + i \sin(n\theta))$ , with applications to trigonometric identities.
- **Complex Numbers and Polynomials:** Understanding how complex numbers arise as polynomial roots, factorization over complex numbers, and the Fundamental Theorem of Algebra.

- **Zeros of a Polynomial:** Finding and interpreting zeros, multiplicity of roots, and relationships between roots and coefficients via Vieta's formulas.
- **Complex Coefficients of a Polynomial:** Working with polynomials that have complex coefficients and understanding when conjugate pairs appear.
- **Vector Problems:** Representing complex numbers as vectors, operations such as addition and scalar multiplication, and applications to geometry including rotations and translations.
- **Roots of Complex Numbers:** Finding  $n$ -th roots using De Moivre's theorem, understanding the geometric distribution of roots on circles, and unity roots.
- **Curves and Regions:** Describing and sketching loci such as circles, lines, rays, and regions in the complex plane defined by equations and inequalities involving modulus and argument.

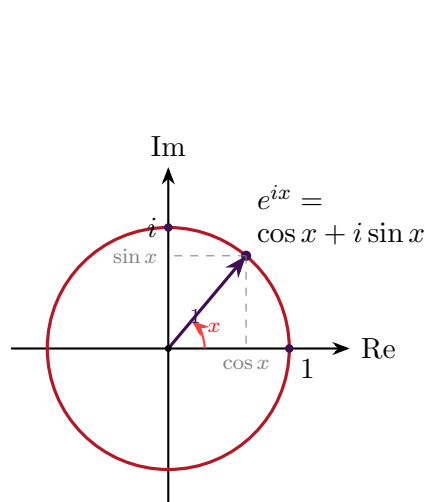
## 1.5 Complex Numbers Primer

### 1.5.1 Key Theorems and Formulas

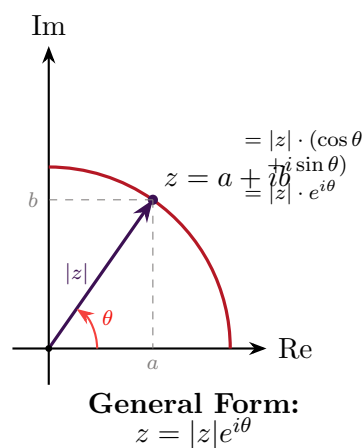
**Euler's Theorem:**

$$e^{i\theta} = \cos \theta + i \sin \theta$$

This connects exponential form with polar form and is fundamental to many proofs.



Unit Circle:  $e^{ix}$



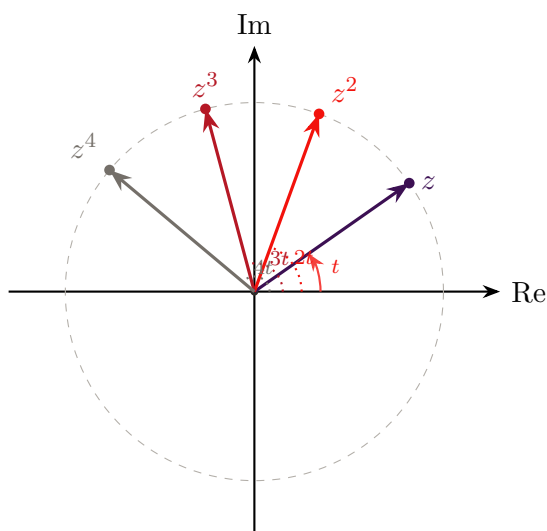
**De Moivre's Theorem:** For any integer  $n$ ,

$$(r(\cos \theta + i \sin \theta))^n = r^n(\cos(n\theta) + i \sin(n\theta))$$

or equivalently,  $(re^{i\theta})^n = r^n e^{in\theta}$ .

This theorem is invaluable for:

- Finding powers of complex numbers
- Finding  $n$ -th roots of complex numbers
- Proving trigonometric identities



For  $z = e^{it}$  on unit circle:

$z^2 = e^{i(2t)}$  rotates by  $2t$

$z^3 = e^{i(3t)}$  rotates by  $3t$

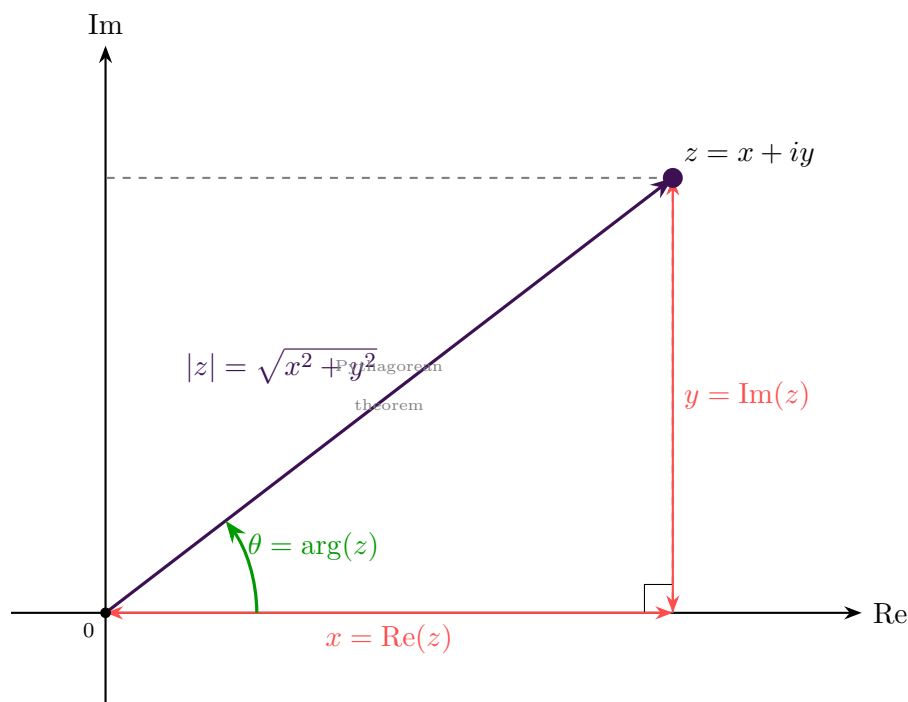
$z^4 = e^{i(4t)}$  rotates by  $4t$

Each power multiplies the angle by  $n$ .

Powers of  $z$  with  $|z| = 1$  and  $\arg(z) = t$

**Modulus and Argument:** For  $z = x + iy$ :

- Modulus:  $|z| = \sqrt{x^2 + y^2}$
- Argument:  $\arg(z) = \tan^{-1}(y/x)$  (with appropriate quadrant adjustments)
- Properties:  $|z_1 z_2| = |z_1| |z_2|$ ,  $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$  (modulo  $2\pi$ )



**Modulus:**

$$|z| = \sqrt{x^2 + y^2}$$

Represents the *distance* from origin to  $z$

**Argument:**

$$\arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

Represents the *angle* from positive real axis

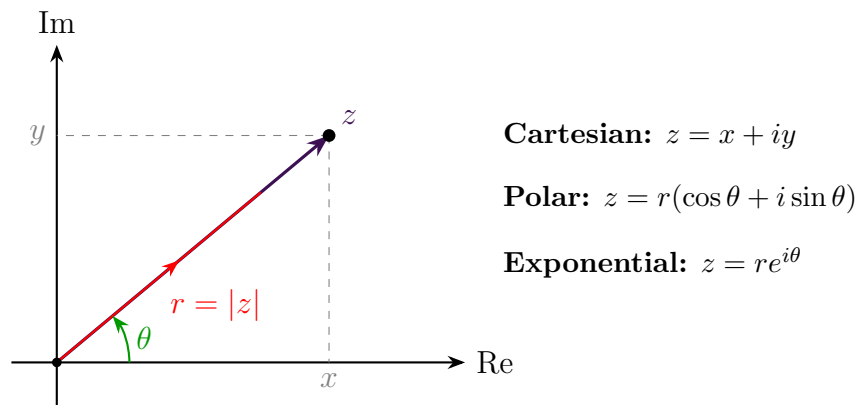
## 1.5.2 Forms of Complex Numbers

**Cartesian Form:**  $z = x + iy$

**Polar Form:**  $z = r(\cos \theta + i \sin \theta)$  where  $r = |z|$  and  $\theta = \arg(z)$

**Exponential Form:**  $z = re^{i\theta}$

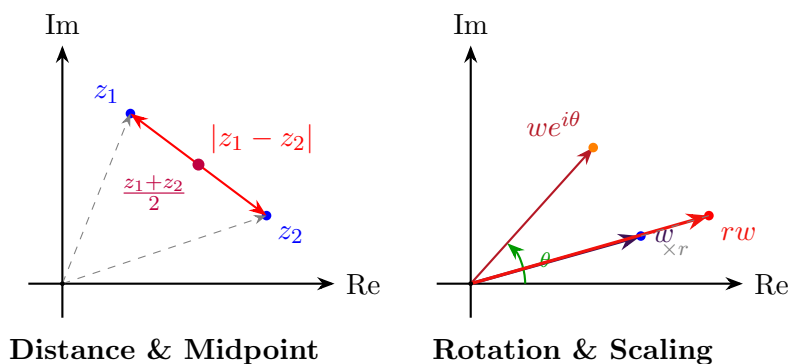
Conversions between these forms are essential skills tested frequently in Extension 2.



### 1.5.3 Argand Diagram

The Argand diagram visualizes complex numbers as points in the plane, with the real part on the horizontal axis and the imaginary part on the vertical axis. Geometric interpretations include:

- $|z_1 - z_2|$  represents the distance between  $z_1$  and  $z_2$
- $(z_1 + z_2)/2$  represents the midpoint
- Multiplication by  $e^{i\theta}$  rotates by angle  $\theta$
- Multiplication by  $r$  scales by factor  $r$



### 1.5.4 2D Rotation with Euler's Equation

The multiplication of two complex numbers implies a rotation in 2D space. When we multiply a complex number  $z$  by  $e^{i\phi}$ , it rotates  $z$  by angle  $\phi$  while preserving its modulus.

**Key Insight:** For complex numbers  $e^{i\theta}$  and  $e^{i\phi}$  on the unit circle:

$$e^{i\theta} \cdot e^{i\phi} = e^{i(\theta+\phi)}$$

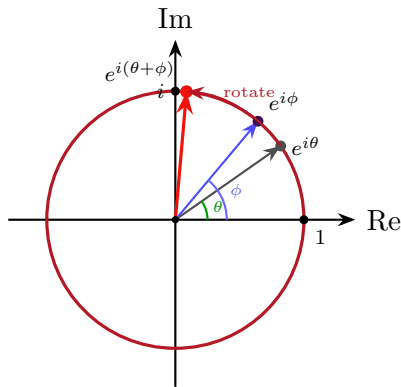
This shows that multiplication adds the angles, which is equivalent to rotation.

**General Rotation:** To rotate any complex number  $z = x + iy = |z|e^{i\theta}$  by angle  $\phi$ :

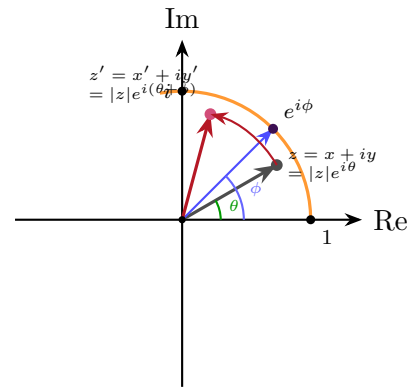
$$\begin{aligned}
 z' &= z \cdot e^{i\phi} = |z|e^{i\theta} \cdot e^{i\phi} \\
 &= |z|e^{i(\theta+\phi)} \\
 &= (x + iy)e^{i\phi} \\
 &= (x + iy)(\cos \phi + i \sin \phi) \\
 &= \cos \phi \cdot x - \sin \phi \cdot y + i(\sin \phi \cdot x + \cos \phi \cdot y)
 \end{aligned}$$

**Matrix Form:** This rotation can be expressed as a 2D rotation matrix:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



Multiplication implies rotation



2D rotation of complex number

**Note:** This concept extends to 3D rotations using *quaternions*, which use a rotation axis vector  $\mathbf{u}$  instead of a single complex number. The quaternion rotation formula is  $\cos \theta + \mathbf{u} \sin \theta$  where  $\mathbf{u}$  is the unit vector along the rotation axis.

### 1.5.5 Notation and Conventions

Throughout this booklet:

- $\bar{z}$  denotes the complex conjugate of  $z$
- $\text{Re}(z)$  and  $\text{Im}(z)$  denote the real and imaginary parts
- $\arg(z)$  denotes the principal argument in  $(-\pi, \pi]$
- Angles are in radians unless otherwise specified

## 2 Part 1: Problems and Solutions (Detailed)

Part 1 contains three sets of problems—basic, medium, and advanced. Each set provides five problems. For every problem we present a comprehensive solution without any hints so that learners focus on the full reasoning trail.

## 2.1 Basic Complex Numbers Problems

### Problem 2.1: Basic Complex Arithmetic

Let  $z = 5 - i$  and  $w = 2 + 3i$ . What is the value of  $2z + \bar{w}$ ?

### Solution 2.1

First, compute  $2z$ :

$$2z = 2(5 - i) = 10 - 2i.$$

Next, find the conjugate of  $w$ :

$$\bar{w} = \overline{2 + 3i} = 2 - 3i.$$

Now add these results:

$$2z + \bar{w} = (10 - 2i) + (2 - 3i) = 12 - 5i.$$

Therefore,  $2z + \bar{w} = 12 - 5i$ .

### Takeaways 2.1

The conjugate of  $a + bi$  is  $a - bi$ . When adding complex numbers, combine real parts with real parts and imaginary parts with imaginary parts.

### Problem 2.2: Finding Square Roots

What value of  $z$  satisfies  $z^2 = 7 - 24i$ ?

### Solution 2.2

Let  $z = a + bi$  where  $a, b$  are real. Then:

$$z^2 = (a + bi)^2 = a^2 - b^2 + 2abi = 7 - 24i.$$

Equating real and imaginary parts:

$$\begin{aligned} a^2 - b^2 &= 7 \\ 2ab &= -24. \end{aligned}$$

From the second equation:  $ab = -12$ , so  $b = -\frac{12}{a}$ .

Substitute into the first equation:

$$a^2 - \left(-\frac{12}{a}\right)^2 = 7 \implies a^2 - \frac{144}{a^2} = 7.$$

Multiply by  $a^2$ :

$$a^4 - 7a^2 - 144 = 0.$$

Let  $u = a^2$ :

$$u^2 - 7u - 144 = 0 \implies (u - 16)(u + 9) = 0.$$

Since  $u = a^2 \geq 0$ , we have  $u = 16$ , so  $a^2 = 16$  giving  $a = \pm 4$ .

If  $a = 4$ :  $b = -\frac{12}{4} = -3$ , so  $z = 4 - 3i$ .

If  $a = -4$ :  $b = -\frac{12}{-4} = 3$ , so  $z = -4 + 3i$ .

Verify:  $(4 - 3i)^2 = 16 - 24i + 9i^2 = 16 - 24i - 9 = 7 - 24i$ . ✓

Therefore,  $z = 4 - 3i$  or  $z = -4 + 3i$ .

**Takeaways 2.2**

To find square roots of complex numbers in Cartesian form, let  $z = a + bi$  and equate real and imaginary parts after expanding  $z^2$ . This gives a system of two equations in two unknowns.

**Problem 2.3: Complex Roots of Quadratics**

Given that  $z = 3 + i$  is a root of  $z^2 + pz + q = 0$ , where  $p$  and  $q$  are real, what are the values of  $p$  and  $q$ ?

**Solution 2.3**

Since the polynomial has real coefficients and  $z = 3 + i$  is a root, its complex conjugate  $\bar{z} = 3 - i$  must also be a root.

By Vieta's formulas:

$$p = -(z_1 + z_2) = -((3 + i) + (3 - i)) = -6,$$

$$q = z_1 \cdot z_2 = (3 + i)(3 - i) = 9 - i^2 = 9 - (-1) = 10.$$

Therefore,  $p = -6$  and  $q = 10$ .

**Takeaways 2.3**

For polynomials with real coefficients, complex roots occur in conjugate pairs. Use Vieta's formulas: sum of roots  $= -p$  and product of roots  $= q$  for  $z^2 + pz + q = 0$ .

**Problem 2.4: Powers of  $i$** 

Write  $i^9$  in the form  $a + ib$  where  $a$  and  $b$  are real.

**Solution 2.4**

Note the pattern of powers of  $i$ :

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = -i$$

$$i^4 = (i^2)^2 = 1$$

$$i^5 = i^4 \cdot i = i$$

The pattern repeats every 4 powers. To find  $i^9$ :

$$9 = 4 \cdot 2 + 1,$$

$$\text{so } i^9 = i^{4 \cdot 2 + 1} = (i^4)^2 \cdot i = 1^2 \cdot i = i.$$

Therefore,  $i^9 = 0 + 1i$ , giving  $a = 0$  and  $b = 1$ .

**Takeaways 2.4**

Powers of  $i$  repeat with period 4:  $i^1 = i$ ,  $i^2 = -1$ ,  $i^3 = -i$ ,  $i^4 = 1$ . Use division by 4 to find  $i^n = i^{n \bmod 4}$ .

**Problem 2.5: Polar Form Conversion**

Write  $1 + i$  in the form  $r(\cos \theta + i \sin \theta)$ .

**Solution 2.5**

For  $z = 1 + i$ , first find the modulus:

$$r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Next, find the argument. Since  $z$  is in the first quadrant:

$$\theta = \arg(z) = \tan^{-1}\left(\frac{1}{1}\right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

Therefore:

$$1 + i = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

**Takeaways 2.5**

To convert  $z = x + iy$  to polar form: (1) Find modulus  $r = \sqrt{x^2 + y^2}$ . (2) Find argument  $\theta = \tan^{-1}(y/x)$  (adjust for quadrant). (3) Write  $z = r(\cos \theta + i \sin \theta)$ .

**2.2 Medium Complex Numbers Problems****Problem 2.6: Rhombus on Argand Diagram**

On the Argand diagram, the complex numbers  $0$ ,  $1 + i\sqrt{3}$ ,  $\sqrt{3} + i$  and  $z$  form a rhombus. Find  $z$  in the form  $a + ib$ .

**Solution 2.6**

In a rhombus, opposite sides are parallel and equal in length. Since we have vertices at  $0$ ,  $1 + i\sqrt{3}$ ,  $\sqrt{3} + i$ , and  $z$ , and they form a rhombus with one vertex at the origin, we need  $z$  to be the fourth vertex. The rhombus has sides:

- From  $0$  to  $1 + i\sqrt{3}$
- From  $0$  to  $\sqrt{3} + i$
- From  $1 + i\sqrt{3}$  to  $z$
- From  $\sqrt{3} + i$  to  $z$

For a rhombus with one vertex at the origin, if two adjacent vertices are at  $w_1$  and  $w_2$ , then the fourth vertex is at  $w_1 + w_2$  (by the parallelogram law).

Therefore:

$$z = (1 + i\sqrt{3}) + (\sqrt{3} + i) = (1 + \sqrt{3}) + i(\sqrt{3} + 1) = (1 + \sqrt{3})(1 + i).$$

Simplifying:

$$z = 1 + \sqrt{3} + i(1 + \sqrt{3}) = (1 + \sqrt{3})(1 + i).$$

Therefore,  $z = (1 + \sqrt{3}) + i(1 + \sqrt{3})$ .

**Takeaways 2.6**

In the Argand diagram, a parallelogram (including rhombus) with vertices at  $0$ ,  $w_1$ ,  $w_2$ , and  $z$  has its fourth vertex at  $z = w_1 + w_2$  by the parallelogram law of vector addition.

**Problem 2.7: De Moivre's Theorem for Real Results**

Given  $-\sqrt{3} - i = 2 \left( \cos \left( -\frac{5\pi}{6} \right) + i \sin \left( -\frac{5\pi}{6} \right) \right)$ , show that  $(-\sqrt{3} - i)^6$  is a real number.

**Solution 2.7**

Using De Moivre's theorem:

$$\begin{aligned} (-\sqrt{3} - i)^6 &= \left[ 2 \left( \cos \left( -\frac{5\pi}{6} \right) + i \sin \left( -\frac{5\pi}{6} \right) \right) \right]^6 \\ &= 2^6 \left( \cos \left( 6 \cdot \left( -\frac{5\pi}{6} \right) \right) + i \sin \left( 6 \cdot \left( -\frac{5\pi}{6} \right) \right) \right) \\ &= 64 (\cos(-5\pi) + i \sin(-5\pi)). \end{aligned}$$

Now evaluate the trigonometric values:

$$\cos(-5\pi) = \cos(5\pi) = \cos(\pi) = -1,$$

$$\sin(-5\pi) = -\sin(5\pi) = -\sin(\pi) = 0.$$

Therefore:

$$(-\sqrt{3} - i)^6 = 64(-1 + 0i) = -64.$$

Since the imaginary part is zero,  $(-\sqrt{3} - i)^6 = -64$  is indeed a real number.

**Takeaways 2.7**

A complex number in polar form  $r(\cos \theta + i \sin \theta)$  raised to power  $n$  has argument  $n\theta$ . If  $n\theta$  is a multiple of  $\pi$ , then  $\sin(n\theta) = 0$  and the result is real.

**Problem 2.8: Division in Polar Form**

Let  $\alpha = 1 + i\sqrt{3}$  and  $\beta = 1 + i$ . Find the modulus-argument form of  $\frac{\alpha}{\beta}$ .

**Solution 2.8**

First, convert  $\alpha$  and  $\beta$  to polar form.

For  $\alpha = 1 + i\sqrt{3}$ :

$$|\alpha| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2,$$

$$\arg(\alpha) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}.$$

So  $\alpha = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ .

For  $\beta = 1 + i$ :

$$|\beta| = \sqrt{1^2 + 1^2} = \sqrt{2},$$

$$\arg(\beta) = \tan^{-1}(1) = \frac{\pi}{4}.$$

So  $\beta = \sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$ .

Now compute the quotient:

$$\begin{aligned}\frac{\alpha}{\beta} &= \frac{2}{\sqrt{2}} \left( \cos \left( \frac{\pi}{3} - \frac{\pi}{4} \right) + i \sin \left( \frac{\pi}{3} - \frac{\pi}{4} \right) \right) \\ &= \sqrt{2} \left( \cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right).\end{aligned}$$

**Takeaways 2.8**

For division in polar form:  $\frac{r_1(\cos \theta_1 + i \sin \theta_1)}{r_2(\cos \theta_2 + i \sin \theta_2)} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$ .

**Problem 2.9: Powers and Cartesian Form**

Express  $(\sqrt{3} - i)^7$  in the form  $x + iy$ .

**Solution 2.9**

First, convert  $\sqrt{3} - i$  to polar form:

$$r = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2,$$

$$\theta = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6} \text{ (fourth quadrant)}.$$

So  $\sqrt{3} - i = 2\left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right)\right)$ .

Apply De Moivre's theorem:

$$\begin{aligned}(\sqrt{3} - i)^7 &= 2^7 \left( \cos \left( 7 \cdot \left( -\frac{\pi}{6} \right) \right) + i \sin \left( 7 \cdot \left( -\frac{\pi}{6} \right) \right) \right) \\ &= 128 \left( \cos \left( -\frac{7\pi}{6} \right) + i \sin \left( -\frac{7\pi}{6} \right) \right).\end{aligned}$$

Now evaluate:

$$\begin{aligned}\cos \left( -\frac{7\pi}{6} \right) &= \cos \left( \frac{7\pi}{6} \right) = \cos \left( \pi + \frac{\pi}{6} \right) = -\cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2}, \\ \sin \left( -\frac{7\pi}{6} \right) &= -\sin \left( \frac{7\pi}{6} \right) = -\sin \left( \pi + \frac{\pi}{6} \right) = -\left( -\sin \frac{\pi}{6} \right) = \frac{1}{2}.\end{aligned}$$

Therefore:

$$(\sqrt{3} - i)^7 = 128 \left( -\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2} \right) = -64\sqrt{3} + 64i.$$

**Takeaways 2.9**

To compute high powers of complex numbers: (1) Convert to polar form. (2) Apply De Moivre's theorem. (3) Simplify the angle using periodicity and reference angles. (4) Convert back to Cartesian form.

**Problem 2.10: Locus as a Curve**

The point  $P$  on the Argand diagram represents  $z = x + iy$  satisfying  $z^2 + \bar{z}^2 = 8$ . Find the equation of the curve in terms of  $x$  and  $y$  and state what type of curve it is.

**Solution 2.10**

Let  $z = x + iy$ , so  $\bar{z} = x - iy$ .

Then:

$$z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy,$$

$$\bar{z}^2 = (x - iy)^2 = x^2 - y^2 - 2ixy.$$

Therefore:

$$z^2 + \bar{z}^2 = (x^2 - y^2 + 2ixy) + (x^2 - y^2 - 2ixy) = 2(x^2 - y^2) = 8.$$

Simplifying:

$$x^2 - y^2 = 4.$$

This can be rewritten as:

$$\frac{x^2}{4} - \frac{y^2}{4} = 1,$$

which is a rectangular hyperbola with center at the origin, with branches along the  $x$ -axis at  $x = \pm 2$ .

**Takeaways 2.10**

To find the locus of points satisfying a complex equation: (1) Substitute  $z = x + iy$  and  $\bar{z} = x - iy$ . (2) Expand and simplify. (3) Equate real and imaginary parts. (4) Identify the curve type.

**2.3 Advanced Complex Numbers Problems****Problem 2.11: Geometric Proof with Complex Numbers**

Points  $A$  and  $B$  represent complex numbers  $z$  and  $w$  on a circle centered at  $O$ . Point  $C$  represents  $z + w$  and also lies on the circle. Show geometrically that  $\angle AOB = \frac{2\pi}{3}$ .

**Solution 2.11**

Given that  $|z| = |w| = |z + w| = r$  (the radius of the circle), we need to find  $\angle AOB$ . Since  $OACB$  forms a parallelogram (by vector addition), and we know:

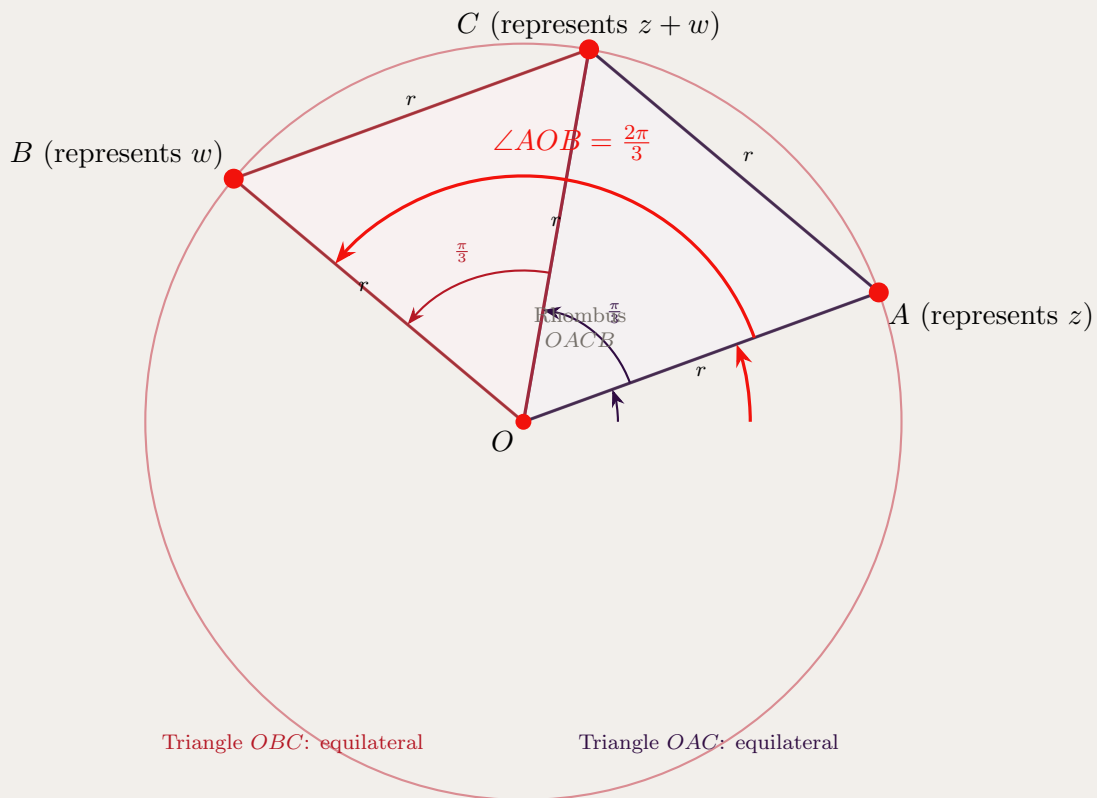
- $OA = |z| = r$
- $OB = |w| = r$
- $OC = |z + w| = r$
- $AC = |w| = r$  (opposite side of parallelogram)
- $BC = |z| = r$  (opposite side of parallelogram)

All sides of the rhombus  $OACB$  have length  $r$ . Moreover, diagonal  $OC$  also has length  $r$ .

Consider triangle  $OAC$ : all three sides have length  $r$  (since  $OA = AC = OC = r$ ), so it is equilateral. Therefore,  $\angle AOC = \frac{\pi}{3}$ .

Similarly, triangle  $OBC$  has  $OB = BC = OC = r$ , so it is also equilateral. Therefore,  $\angle BOC = \frac{\pi}{3}$ . Thus:

$$\angle AOB = \angle AOC + \angle BOC = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}.$$

**Takeaways 2.11**

When complex numbers are represented geometrically, vector addition corresponds to the parallelogram law. A rhombus with all sides equal to its diagonal consists of two equilateral triangles.

**Problem 2.12: Complex Division and Real Parts**

Let  $z = 2(\cos \theta + i \sin \theta)$ . Show that the real part of  $\frac{1}{1-z}$  is  $\frac{1-2\cos \theta}{5-4\cos \theta}$ .

**Solution 2.12**

First, compute  $1 - z$ :

$$1 - z = 1 - 2(\cos \theta + i \sin \theta) = (1 - 2\cos \theta) - 2i \sin \theta.$$

To find  $\frac{1}{1-z}$ , multiply numerator and denominator by the conjugate of  $1 - z$ :

$$\begin{aligned} \frac{1}{1-z} &= \frac{1}{(1-2\cos \theta) - 2i \sin \theta} \cdot \frac{(1-2\cos \theta) + 2i \sin \theta}{(1-2\cos \theta) + 2i \sin \theta} \\ &= \frac{(1-2\cos \theta) + 2i \sin \theta}{(1-2\cos \theta)^2 + (2\sin \theta)^2}. \end{aligned}$$

Compute the denominator:

$$\begin{aligned} (1-2\cos \theta)^2 + 4\sin^2 \theta &= 1 - 4\cos \theta + 4\cos^2 \theta + 4\sin^2 \theta \\ &= 1 - 4\cos \theta + 4(\cos^2 \theta + \sin^2 \theta) \\ &= 1 - 4\cos \theta + 4 \\ &= 5 - 4\cos \theta. \end{aligned}$$

Therefore:

$$\frac{1}{1-z} = \frac{(1-2\cos \theta) + 2i \sin \theta}{5-4\cos \theta}.$$

The real part is:

$$\operatorname{Re} \left( \frac{1}{1-z} \right) = \frac{1-2\cos \theta}{5-4\cos \theta}.$$

**Takeaways 2.12**

To find the real part of a complex fraction, rationalize by multiplying by the conjugate of the denominator. Use the identity  $\cos^2 \theta + \sin^2 \theta = 1$  to simplify.

**Problem 2.13: Finding Complex Roots of Polynomials**

Two zeros of  $P(x) = x^4 - 12x^3 + 59x^2 - 138x + 130$  are  $a + ib$  and  $a + 2ib$  where  $a, b$  are real and  $b > 0$ . Find  $a$  and  $b$ .

**Solution 2.13**

Since  $P(x)$  has real coefficients, complex roots occur in conjugate pairs. If  $a + ib$  is a root, then  $a - ib$  is also a root. Similarly, if  $a + 2ib$  is a root, then  $a - 2ib$  is also a root.

Therefore, the four roots are:  $a + ib$ ,  $a - ib$ ,  $a + 2ib$ ,  $a - 2ib$ .

By Vieta's formulas, the sum of roots equals the negative of the coefficient of  $x^3$  divided by the leading coefficient:

$$(a + ib) + (a - ib) + (a + 2ib) + (a - 2ib) = 4a = 12,$$

so  $a = 3$ .

The product of roots equals the constant term:

$$[(a + ib)(a - ib)][(a + 2ib)(a - 2ib)] = (a^2 + b^2)(a^2 + 4b^2) = 130.$$

Substituting  $a = 3$ :

$$(9 + b^2)(9 + 4b^2) = 130.$$

Expand:

$$81 + 36b^2 + 9b^2 + 4b^4 = 130,$$

$$4b^4 + 45b^2 + 81 = 130,$$

$$4b^4 + 45b^2 - 49 = 0.$$

Let  $u = b^2$ :

$$4u^2 + 45u - 49 = 0.$$

Using the quadratic formula:

$$u = \frac{-45 \pm \sqrt{45^2 + 4 \cdot 4 \cdot 49}}{8} = \frac{-45 \pm \sqrt{2025 + 784}}{8} = \frac{-45 \pm \sqrt{2809}}{8} = \frac{-45 \pm 53}{8}.$$

Since  $u = b^2 \geq 0$ :  $u = \frac{-45+53}{8} = \frac{8}{8} = 1$ .

Therefore,  $b^2 = 1$ , so  $b = 1$  (since  $b > 0$ ).

Thus,  $a = 3$  and  $b = 1$ .

**Takeaways 2.13**

For polynomials with real coefficients, complex roots come in conjugate pairs. Use Vieta's formulas (sum and product of roots) along with conjugate pairing to set up equations for unknown parameters.

**Problem 2.14: Trigonometric Identity via Complex Numbers**

Show that  $\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$ .

**Solution 2.14**

By De Moivre's theorem:

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta.$$

Expand the left side using the binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \sum_{k=0}^4 \binom{4}{k} \cos^{4-k} \theta (i \sin \theta)^k \\ &= \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6i^2 \cos^2 \theta \sin^2 \theta \\ &\quad + 4i^3 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta \\ &= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta \\ &\quad - 4i \cos \theta \sin^3 \theta + \sin^4 \theta \\ &= (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) \\ &\quad + i(4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta). \end{aligned}$$

Equating real parts:

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta.$$

**Takeaways 2.14**

De Moivre's theorem combined with binomial expansion provides a systematic way to derive multiple-angle formulas. Equate real and imaginary parts to obtain separate identities for  $\cos(n\theta)$  and  $\sin(n\theta)$ .

**Problem 2.15: Conjugate Pairs and De Moivre**

Show that

$$(1 + i)^n + (1 - i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}$$

.

**Solution 2.15**

First, convert  $1 + i$  and  $1 - i$  to polar form.

For  $1 + i$ :

$$|1 + i| = \sqrt{2}, \quad \arg(1 + i) = \frac{\pi}{4},$$

$$\text{so } 1 + i = \sqrt{2} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = \sqrt{2} e^{i\pi/4}.$$

For  $1 - i$ :

$$|1 - i| = \sqrt{2}, \quad \arg(1 - i) = -\frac{\pi}{4},$$

$$\text{so } 1 - i = \sqrt{2} \left( \cos \left( -\frac{\pi}{4} \right) + i \sin \left( -\frac{\pi}{4} \right) \right) = \sqrt{2} e^{-i\pi/4}.$$

By De Moivre's theorem:

$$(1 + i)^n = (\sqrt{2})^n e^{in\pi/4} = (\sqrt{2})^n \left( \cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right),$$

$$(1 - i)^n = (\sqrt{2})^n e^{-in\pi/4} = (\sqrt{2})^n \left( \cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right).$$

Adding:

$$\begin{aligned} (1 + i)^n + (1 - i)^n &= (\sqrt{2})^n \left[ \left( \cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right) + \left( \cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right) \right] \\ &= (\sqrt{2})^n \cdot 2 \cos \frac{n\pi}{4} \\ &= 2(\sqrt{2})^n \cos \frac{n\pi}{4}. \end{aligned}$$

**Takeaways 2.15**

For conjugate pairs  $z$  and  $\bar{z}$  in polar form,  $z^n + \bar{z}^n = 2|z|^n \cos(n \arg(z))$  because the imaginary parts cancel. This is useful for simplifying sums involving conjugate powers.

### 3 Part 2: Problems and Solutions (Concise + Hints)

Part 2 revisits three new sets of problems. Solutions are intentionally briefer to encourage student ownership, and every problem includes an upside-down hint.

#### 3.1 Basic Complex Numbers Problems

**Problem 3.1: Multiplication by  $i$  as Rotation**

Consider the complex number  $z = 3 + 2i$  on the Argand diagram. Find the complex number  $iz$  and describe its geometric relationship to  $z$ .

**Hint:** Multiplication by  $i$  rotates a complex number by  $90^\circ$  counterclockwise about the origin while preserving its modulus. If  $z = a + bi$ , calculate  $iz$  algebraically.

**Solution 3.1: Sketch****Algebraic calculation:**

Given  $z = 3 + 2i$ , we compute:

$$\begin{aligned} iz &= i(3 + 2i) \\ &= 3i + 2i^2 \\ &= 3i + 2(-1) \\ &= -2 + 3i \end{aligned}$$

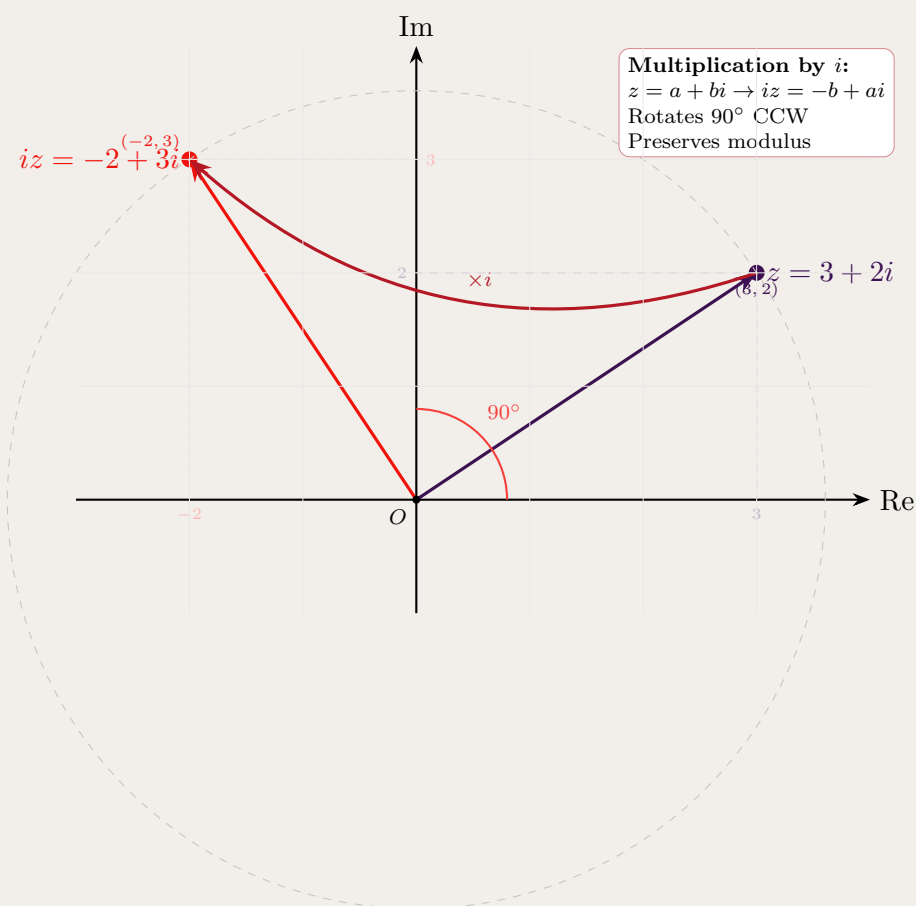
Therefore:  $iz = -2 + 3i$

**General rule:** If  $z = a + bi$ , then:

$$iz = i(a + bi) = ai + bi^2 = -b + ai$$

This shows that  $(a, b) \rightarrow (-b, a)$ , which is a  $90^\circ$  counterclockwise rotation.

**Verification for our example:** - Original point:  $z = 3 + 2i$  at position  $(3, 2)$  - Rotated point:  $iz = -2 + 3i$  at position  $(-2, 3)$  - Check:  $(3, 2) \rightarrow (-2, 3)$  is correct

**Key observations:**

- The modulus is preserved:  $|z| = |iz| = \sqrt{13}$
- The transformation  $(a, b) \rightarrow (-b, a)$  is a  $90^\circ$  counterclockwise rotation
- Both points lie on the same circle centered at the origin
- This geometric property makes  $i$  rotation in the complex plane

**Problem 3.2: Quadrant Determination**

The Argand diagram shows  $z$  in the first quadrant and  $w$  in the second quadrant. Which complex number could lie in the 3rd quadrant:  $-w$ ,  $2iz$ ,  $\bar{z}$ , or  $w - z$ ?

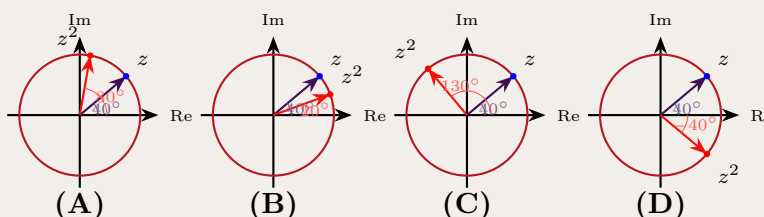
**Hint:** Check the signs of the real and imaginary parts of each option. Third quadrant means both parts are negative.

**Solution 3.2: Sketch**

If  $z$  is in Q1:  $\operatorname{Re}(z) > 0, \operatorname{Im}(z) > 0$ . If  $w$  is in Q2:  $\operatorname{Re}(w) < 0, \operatorname{Im}(w) > 0$ . Then  $w - z$  has  $\operatorname{Re}(w - z) = \operatorname{Re}(w) - \operatorname{Re}(z) < 0$  and  $\operatorname{Im}(w - z) = \operatorname{Im}(w) - \operatorname{Im}(z)$  which could be negative if  $\operatorname{Im}(z) > \operatorname{Im}(w)$ . Answer: (D)  $w - z$ .

**Problem 3.3: Doubling the Argument**

A complex number  $z$  is on the unit circle at angle  $\theta = 40^\circ$ . Which diagram best shows the location of  $\frac{z^2}{|z|}$ ?



**Hint:** If  $z = e^{i\theta}$ , then  $z^2 = e^{i2\theta}$  and  $|z| = 1$ .

**Solution 3.3: Sketch**

Since  $|z| = 1$ , we have  $\frac{z^2}{|z|} = z^2$ . By De Moivre's theorem,  $z^2$  has the same modulus but double the argument. If  $z$  is at angle  $\theta = 40^\circ$ , then  $z^2$  is at angle  $2\theta = 80^\circ$ .

**Analysis of options:**

- (A) **CORRECT:** Shows  $z^2$  at angle  $80^\circ = 2 \times 40^\circ$ . This correctly applies De Moivre's theorem.
- (B) Incorrect: Shows  $z^2$  at angle  $20^\circ = 40^\circ/2$ . This halves the angle instead of doubling it.
- (C) Incorrect: Shows  $z^2$  at angle  $130^\circ = 40^\circ + 90^\circ$ . This adds a fixed angle, which would represent multiplication by  $i$ .
- (D) Incorrect: Shows  $z^2$  at angle  $-40^\circ$ . This reflects across the real axis, which would give the conjugate  $\bar{z}$ .

**Answer: (A)**

**Problem 3.4: Polynomial with Complex Zero**

Which polynomial could have  $2 + i$  as a zero, given that  $k$  is real?

- (A)  $x^3 - 4x^2 + kx$   
 (B)  $x^3 - 4x^2 + kx + 5$   
 (C)  $x^3 - 5x^2 + kx$   
 (D)  $x^3 - 5x^2 + kx + 5$

**Hint:** If the polynomial has real coefficients and  $2 + i$  is a root, then  $2 - i$  must also be a root. The sum of these roots is 4.

**Solution 3.4: Sketch**

Roots  $2 + i$  and  $2 - i$  sum to 4. Let the third root be  $r$ . Then sum of all roots  $= 4 + r$ . From the coefficient of  $x^2$ :  $-(4 + r) = -4$  or  $-5$ . If  $-4$ , then  $r = 0$ , giving option (A) or (B). Check:  $(x)(x^2 - 4x + 5) = x^3 - 4x^2 + 5x$ , but we need  $+5$  constant. Answer: (B).

**Problem 3.5: Squaring Complex Numbers**

Let  $z = 3 + i$  and  $w = 2 - 5i$ . Find  $z^2$  in the form  $x + iy$ .

**Hint:** Use  $(a + bi)^2 = a^2 - b^2 + 2abi$ .

**Solution 3.5: Sketch**

$$z^2 = (3 + i)^2 = 9 + 6i + i^2 = 9 + 6i - 1 = 8 + 6i.$$

**Problem 3.6: Conjugate Subtraction**

Let  $z = 4 + i$  and  $w = \bar{z}$ . Find  $w - z$  in the form  $x + iy$ .

**Hint:** If  $z = 4 + i$ , then  $\bar{z} = 4 - i$ .

**Solution 3.6: Sketch**

$$w = \bar{z} = 4 - i. \text{ Then } w - z = (4 - i) - (4 + i) = -2i.$$

**Problem 3.7: Division by Complex Number**

Let  $z = 2 + i$ . Find  $\frac{4}{z}$  in the form  $x + iy$ .

**Hint:** Multiply numerator and denominator by  $\bar{z} = 2 - i$ .

**Solution 3.7: Sketch**

$$\frac{4}{2+i} = \frac{4(2-i)}{(2+i)(2-i)} = \frac{8-4i}{4-i^2} = \frac{8-4i}{5} = \frac{8}{5} - \frac{4}{5}i.$$

**Problem 3.8: Complex Multiplication**

Let  $z = 3 + i$  and  $w = 1 - i$ . Find  $zw$  in the form  $x + iy$ .

**Hint:** Use FOIL:  $(a + bi)(c + di) = ac + bd + (ad + bc)i$ .

**Solution 3.8: Sketch**

$$zw = (3 + i)(1 - i) = 3 - 3i + i - i^2 = 3 - 2i + 1 = 4 - 2i.$$

**Problem 3.9: Another Division**

Let  $w = 1 - i$ . Find  $\frac{6}{w}$  in the form  $x + iy$ .

**Hint:** Multiply by conjugate:  $\bar{w} = 1 + i$ .

**Solution 3.9: Sketch**

$$\frac{6}{1-i} = \frac{6(1+i)}{(1-i)(1+i)} = \frac{6+6i}{1-i^2} = \frac{6+6i}{2} = 3 + 3i.$$

**Problem 3.10: Dividing by Conjugate**

Let  $z = 4 + i$  and  $w = \bar{z}$ . Find  $\frac{z}{w}$  in the form  $x + iy$ .

**Hint:**  $w = 4 - i$ , so  $\frac{z}{w} = \frac{4+i}{4-i}$ .

**Solution 3.10: Sketch**

$$\frac{4+i}{4-i} = \frac{(4+i)(4+i)}{(4-i)(4+i)} = \frac{16+8i+i^2}{16-i^2} = \frac{15+8i}{17} = \frac{15}{17} + \frac{8}{17}i.$$

**Problem 3.11: Modulus from Polar Form**

Let  $z = \frac{1}{2}(\cos \theta + i \sin \theta)$ . Find  $|z|$ .

**Hint:** For  $z = r(\cos \theta + i \sin \theta)$ , we have  $|z| = r$ .

**Solution 3.11: Sketch**

$$|z| = \frac{1}{2}.$$

**Problem 3.12: Division in Cartesian Form**

Express  $\frac{3-i}{2+i}$  in the form  $x + iy$ .

Hint: Multiply by  $\frac{2-i}{2-i}$ .

**Solution 3.12: Sketch**

$$\frac{3-i}{2+i} = \frac{(3-i)(2-i)}{(2+i)(2-i)} = \frac{6-3i-2i+i^2}{4-i^2} = \frac{5-5i}{5} = 1-i.$$

**Problem 3.13: Addition with Conjugate**

Let  $z = 1 + 3i$  and  $w = 2 - i$ . Find  $z + \bar{w}$ .

Hint:  $\bar{w} = 2 + i$ .

**Solution 3.13: Sketch**

$$z + \bar{w} = (1 + 3i) + (2 + i) = 3 + 4i.$$

**Problem 3.14: Product with Conjugate**

Evaluate  $w\bar{z}$  given  $w = -1 + 4i$  and  $z = 2 - i$ .

Hint:  $\bar{z} = 2 + i$ .

**Solution 3.14: Sketch**

$$w\bar{z} = (-1 + 4i)(2 + i) = -2 - i + 8i + 4i^2 = -2 + 7i - 4 = -6 + 7i.$$

**Problem 3.15: Polar Form of Given Number**

Let  $w = 1 + i\sqrt{3}$ . Express  $w$  in modulus-argument form.

Hint:  $|w| = \sqrt{1+3} = 2$ , and  $\arg(w) = \tan^{-1}(\sqrt{3}/1) = \pi/3$ .

**Solution 3.15: Sketch**

$$w = 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right).$$

**Problem 3.16: Showing a Power is Real**

Let  $z = \sqrt{3} - i$ . Show that  $z^6$  is real.

**Hint:** Convert to polar form:  $z = 2(\cos(-\pi/6) + i \sin(-\pi/6))$ .

**Solution 3.16: Sketch**

$$z^6 = 2^6(\cos(-\pi) + i \sin(-\pi)) = 64(-1 + 0i) = -64, \text{ which is real.}$$

**Problem 3.17: Principal Argument**

Evaluate  $\text{Arg}(z)$  for  $z = -\sqrt{3} + i$ .

**Hint:**  $z$  is in the second quadrant. Use  $\tan \theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$ .

**Solution 3.17: Sketch**

Reference angle is  $\pi/6$ . Since  $z$  is in Q2,  $\text{Arg}(z) = \pi - \pi/6 = \frac{5\pi}{6}$ .

**Problem 3.18: Polynomial with Complex Root**

It is given that  $z = 2 + i$  is a root of  $z^3 + az^2 - 7z + 15 = 0$ , where  $a$  is a real number. What is the value of  $a$ ?

- (A)  $-1$
- (B)  $1$
- (C)  $7$
- (D)  $-7$

**Hint:** Since the polynomial has real coefficients and  $2 + i$  is a root, then  $2 - i$  must also be a root. Use Vieta's formulas or substitute the root directly.

**Solution 3.18: Sketch****Method 1: Direct substitution**

Since  $z = 2 + i$  is a root, substitute into the equation:

$$(2 + i)^3 + a(2 + i)^2 - 7(2 + i) + 15 = 0$$

Calculate  $(2 + i)^2$ :

$$(2 + i)^2 = 4 + 4i + i^2 = 4 + 4i - 1 = 3 + 4i$$

Calculate  $(2 + i)^3 = (2 + i)(3 + 4i)$ :

$$(2 + i)(3 + 4i) = 6 + 8i + 3i + 4i^2 = 6 + 11i - 4 = 2 + 11i$$

Substitute into the equation:

$$\begin{aligned} (2 + 11i) + a(3 + 4i) - 7(2 + i) + 15 &= 0 \\ (2 + 11i) + (3a + 4ai) + (-14 - 7i) + 15 &= 0 \\ (2 + 3a - 14 + 15) + (11 + 4a - 7)i &= 0 \\ (3 + 3a) + (4 + 4a)i &= 0 \end{aligned}$$

For this to equal zero, both real and imaginary parts must be zero:

$$\text{Real part: } 3 + 3a = 0 \implies a = -1$$

$$\text{Imaginary part: } 4 + 4a = 0 \implies a = -1$$

**Method 2: Using conjugate roots**

Since the polynomial has real coefficients, if  $2 + i$  is a root, then  $2 - i$  is also a root.

Let the third root be  $r$ . By Vieta's formulas, the sum of roots equals  $-a$ :

$$(2 + i) + (2 - i) + r = -a$$

$$4 + r = -a$$

The product of roots equals  $-15$ :

$$(2 + i)(2 - i) \cdot r = -15$$

$$(4 - i^2) \cdot r = -15$$

$$5r = -15$$

$$r = -3$$

Therefore:  $4 + (-3) = -a \implies a = -1$

**Answer: (A)  $a = -1$**

**Problem 3.19: Power Using De Moivre's Theorem**

- (i) Write the number  $\sqrt{3} + i$  in modulus-argument form.
- (ii) Hence, or otherwise, write  $(\sqrt{3} + i)^7$  in exact Cartesian form.

**Hint:** For part (i): Find  $|z| = \sqrt{3+1} = 2$  and  $\arg(z) = \tan^{-1}(1/\sqrt{3}) = \pi/6$ . For part (ii): Use De Moivre's theorem to find  $(re^{i\theta})^7 = r^7 e^{i7\theta}$ .

**Solution 3.19: Sketch****Part (i): Convert to modulus-argument form**

Let  $z = \sqrt{3} + i$ .

Calculate the modulus:

$$|z| = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3+1} = \sqrt{4} = 2$$

Calculate the argument. Since  $z$  is in the first quadrant:

$$\arg(z) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

Therefore, in modulus-argument form:

$$z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 2e^{i\pi/6}$$

**Part (ii): Calculate  $z^7$  using De Moivre's theorem**

By De Moivre's theorem:

$$\begin{aligned} z^7 &= \left[2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right]^7 \\ &= 2^7 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right) \\ &= 128 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right) \end{aligned}$$

Calculate the trigonometric values. The angle  $\frac{7\pi}{6}$  is in the third quadrant (reference angle  $\frac{\pi}{6}$ ):

$$\cos \frac{7\pi}{6} = -\cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2}$$

$$\sin \frac{7\pi}{6} = -\sin \frac{\pi}{6} = -\frac{1}{2}$$

Therefore:

$$\begin{aligned} z^7 &= 128 \left(-\frac{\sqrt{3}}{2} + i\left(-\frac{1}{2}\right)\right) \\ &= 128 \left(-\frac{\sqrt{3}}{2} - \frac{i}{2}\right) \\ &= -64\sqrt{3} - 64i \end{aligned}$$

**Answer:**  $(\sqrt{3} + i)^7 = -64\sqrt{3} - 64i$

**Verification (optional):** We can verify that  $|z^7| = |z|^7 = 2^7 = 128$  and  $\arg(z^7) = 7\arg(z) = 7 \cdot \frac{\pi}{6} = \frac{7\pi}{6}$  (in the range  $(-\pi, \pi]$ ).

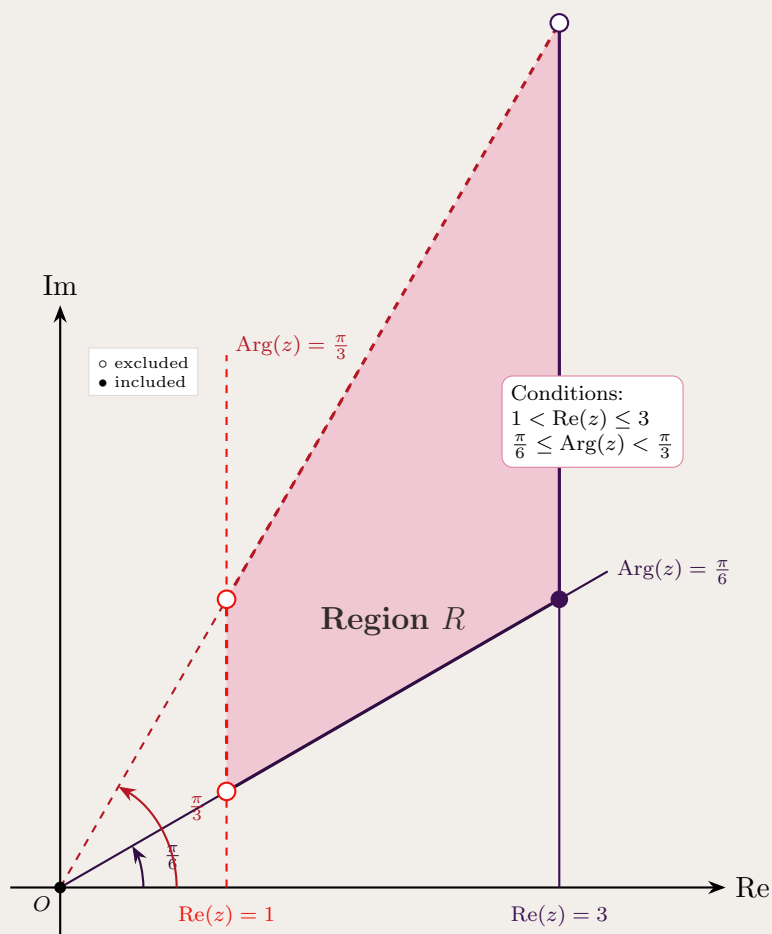
**3.2 Medium Complex Numbers Problems****Problem 3.20: Region with Real and Argument Constraints**

Let  $R$  be the region in the complex plane defined by  $1 < \operatorname{Re}(z) \leq 3$  and  $\frac{\pi}{6} \leq \operatorname{Arg}(z) < \frac{\pi}{3}$ . Sketch this region.

**Hint:** The region is bounded by two vertical lines ( $x = 1$  and  $x = 3$ ) and two rays from the origin at angles  $\pi/6$  and  $\pi/3$ .

**Solution 3.20: Sketch**

Draw vertical lines at  $x = 1$  (dashed, excluded) and  $x = 3$  (solid, included). Draw rays from origin at angles  $30^\circ$  (solid) and  $60^\circ$  (dashed). The region is the intersection in the first quadrant.



**Problem 3.21: The Spira Mirabilis - Eadem mutata resurgo**

*"Eadem mutata resurgo"* — "Changed, I rise again the same."

This phrase was chosen by mathematician Jacob Bernoulli to describe the Logarithmic Spiral, a curve that retains its shape as it scales.

In this problem, you will explore the properties of a (discrete) logarithmic spiral defined in the complex plane, which exhibits self-similarity and convergent path length, a pattern often seen in nature such as in shells, storm systems, and galaxies.

Let  $\{z_n\}$  be a sequence of complex numbers defined by the initial value  $z_0 = 1$  and the recurrence relation:

$$z_{n+1} = \left(\frac{1+i}{2}\right) z_n$$

- (i) Prove that for all  $n \geq 0$ , the triangle formed by the origin  $O$  and two consecutive terms,  $\triangle Oz_n z_{n+1}$ , is similar to the triangle  $\triangle Oz_{n+1} z_{n+2}$ .
- (ii) Find the length of the polygonal spiral path connecting the points  $z_0, z_1, z_2, \dots$ :

$$L = \sum_{n=0}^{\infty} |z_{n+1} - z_n|$$

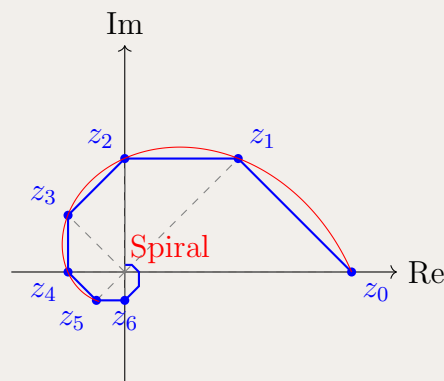
- (iii) Show that the points  $z_n$  lie on the continuous logarithmic spiral defined by the polar equation:

$$r = e^{-k\theta}$$

and find the exact value of the constant  $k$ .

- (iv) Let  $R = |z_0|$ . Prove that the ratio of the total path length to the initial radius is given by:

$$\frac{L}{R} = \cot\left(\frac{\pi}{8}\right)$$



**Hint:**

- **Part (i):** Use the properties of complex multiplication. If  $z_{n+1} = wz_n$ , then multiplication by  $w$  scales the modulus by  $|w|$  and rotates the argument by  $\arg(w)$ . Compare the ratios of sides and included angles for the two triangles.
- **Part (ii):** The sequence of segment lengths forms a geometric progression. First, find  $|z_{n+1} - z_n|$  in terms of  $|z_n|$  and the constant multiplier.
- **Part (iii):** Express the multiplier in exponential form  $re^{i\phi}$ . This allows you to write  $z_n$  as a function of  $n$ . Eliminate  $n$  to relate the modulus (radius) directly to the argument (angle).
- **Part (iv):** Calculate  $\cot(\pi/8)$  exactly using half-angle formulae and compare it to your result from Part (ii).

**Solution 3.21: Detailed**

Let  $w = \frac{1+i}{2}$ . Then  $|w| = \frac{|1+i|}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$  and  $\arg(w) = \frac{\pi}{4}$ .

**(i) Self-similarity.** Since  $z_{n+1} = wz_n$ , multiplication by  $w$  maps the vector  $Oz_n$  to  $Oz_{n+1}$  by scaling all lengths by  $|w|$  and rotating all directions by  $\arg(w)$ . Hence both the included angle  $\angle z_n Oz_{n+1}$  and the ratio  $\frac{|Oz_{n+1}|}{|Oz_n|}$  are constant and equal to  $\arg(w)$  and  $|w|$ , respectively. By SAS,  $\triangle Oz_n z_{n+1} \sim \triangle Oz_{n+1} z_{n+2}$ .

**(ii) Total polygonal length.** The  $n$ -th segment length is

$$\ell_n = |z_{n+1} - z_n| = |wz_n - z_n| = |z_n| |w - 1|.$$

With  $|z_n| = |w|^n = (1/\sqrt{2})^n$  and

$$|w - 1| = \left| \frac{1+i}{2} - 1 \right| = \left| -\frac{1}{2} + \frac{i}{2} \right| = \frac{1}{\sqrt{2}},$$

we get  $\ell_n = (1/\sqrt{2})^{n+1}$ . Therefore

$$L = \sum_{n=0}^{\infty} \ell_n = \sum_{n=0}^{\infty} \left( \frac{1}{\sqrt{2}} \right)^{n+1} = \frac{\frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1.$$

**(iii) Continuous spiral form.** Write  $w = \frac{1}{\sqrt{2}}e^{i\pi/4}$ . Then

$$z_n = w^n = \left( \frac{1}{\sqrt{2}} \right)^n e^{in\pi/4},$$

so in polar form  $r_n = 2^{-n/2}$  and  $\theta_n = \frac{n\pi}{4}$ . Eliminating  $n$  via  $n = \frac{4\theta}{\pi}$  gives

$$r = 2^{-\frac{1}{2} \cdot \frac{4\theta}{\pi}} = 2^{-\frac{2\theta}{\pi}} = e^{-\frac{2\ln 2}{\pi} \theta},$$

hence  $r = e^{-k\theta}$  with  $k = \frac{2\ln 2}{\pi}$ .

**(iv) Ratio as a trig value.** With  $R = |z_0| = 1$ , we have  $\frac{L}{R} = \sqrt{2} + 1$ . To link this to  $\cot(\frac{\pi}{8})$ , set  $t = \tan(\frac{\pi}{8})$  and use the double-angle identity

$$\tan(2A) = \frac{2 \tan A}{1 - \tan^2 A}, \quad A = \frac{\pi}{8},$$

together with  $\tan(\frac{\pi}{4}) = 1$ . Then

$$1 = \frac{2t}{1 - t^2} \implies t^2 + 2t - 1 = 0 \implies t = \sqrt{2} - 1 \quad (\text{positive since } \frac{\pi}{8} \in (0, \frac{\pi}{4})).$$

Thus

$$\cot\left(\frac{\pi}{8}\right) = \frac{1}{\tan(\frac{\pi}{8})} = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 = \frac{L}{R}.$$

## Takeaways

- **Complex Multiplication is Rotation/Scaling:** This problem visualizes the geometric effect of multiplying by a complex number.  $z_{n+1} = wz_n$  rotates the point by  $\arg(w)$  and scales it by  $|w|$ , creating a discrete spiral.
- **Infinite Geometric Series:** The "length of a spiral" is a classic application of summing a geometric series to infinity. Even though the spiral winds infinitely many times, the total distance traveled is finite.
- **Discrete vs Continuous:** The problem bridges the gap between discrete sequences (points  $z_n$ ) and continuous calculus concepts (the smooth logarithmic spiral curve).
- The logarithmic spiral appears widely in nature and has various scientific and engineering applications such as animal shells, galaxies, and antenna designs.
- What is the transformation when multiplying by a complex number. In this case:  $w = \frac{1+i}{2}$ . How does this relate to geometric shapes and patterns in the complex plane?

### Problem 3.22: Statement Analysis

Which statement about complex numbers is true?

- (A) For  $z = x + iy$ , the argument is given by  $\arg(z) = \arctan(y/x)$  for all  $z \neq 0$ .
- (B) For any two complex numbers  $z_1$  and  $z_2$ , we have  $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$ .
- (C) If two complex numbers have the same modulus and their arguments satisfy  $\theta_1 = \theta_2 + 2\pi k$  for some integer  $k$ , then they are equal (modulo  $2\pi$ ).
- (D) For  $z = x + iy$  with  $x > 0$ , the argument is  $\arg(z) = \arctan(y/x)$ .

**Hint:** Check each statement. Remember that  $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$  only modulo  $2\pi$ .

### Solution 3.22: Sketch

- **(A) False:** The formula  $\arg(z) = \arctan(y/x)$  doesn't account for the quadrant. For example, if  $z = -1 + i$ , then  $\arctan(-1) = -\pi/4$ , but the actual argument is  $3\pi/4$  (in the second quadrant).
- **(B) False:** Principal argument addition requires modulo  $2\pi$  adjustment. For instance, if  $\text{Arg}(z_1) = 3\pi/4$  and  $\text{Arg}(z_2) = 3\pi/4$ , then  $\text{Arg}(z_1) + \text{Arg}(z_2) = 3\pi/2$ , but we need to adjust to the principal range  $(-\pi, \pi]$ , giving  $\text{Arg}(z_1 z_2) = -\pi/2$ .
- **(C) True (with proper interpretation):** If  $|z_1| = |z_2|$  and  $\arg(z_1) = \arg(z_2) + 2\pi k$  for integer  $k$ , then  $z_1 = z_2$  since arguments that differ by  $2\pi k$  represent the same direction.
- **(D) True (but limited):** This is correct when  $x > 0$  (first and fourth quadrants), as the  $\arctan$  function gives the correct argument in these quadrants without adjustment.

**Answer:** The best answer is **(D)** for its specific domain, though **(C)** is also correct if "equality" means modulo  $2\pi$ . Statement **(D)** is the most straightforward true statement.

**Problem 3.23: Roots of Unity Application**

Suppose that  $x + \frac{1}{x} = -1$ . What is the value of  $x^{2028} + \frac{1}{x^{2028}}$ ?

**Hint:** Solve  $x^2 + x + 1 = 0$  to find  $x = e^{i2\pi/3}$  or  $x = e^{-i2\pi/3}$ . These are cube roots of unity.

**Solution 3.23: Sketch**

$x^2 + x + 1 = 0$  gives  $x = e^{\pm i2\pi/3}$ . Since  $x^3 = 1$ , we have  $x^{2028} = x^{3 \cdot 676} = 1$ . Thus  $x^{2028} + \frac{1}{x^{2028}} = 1 + 1 = 2$ .

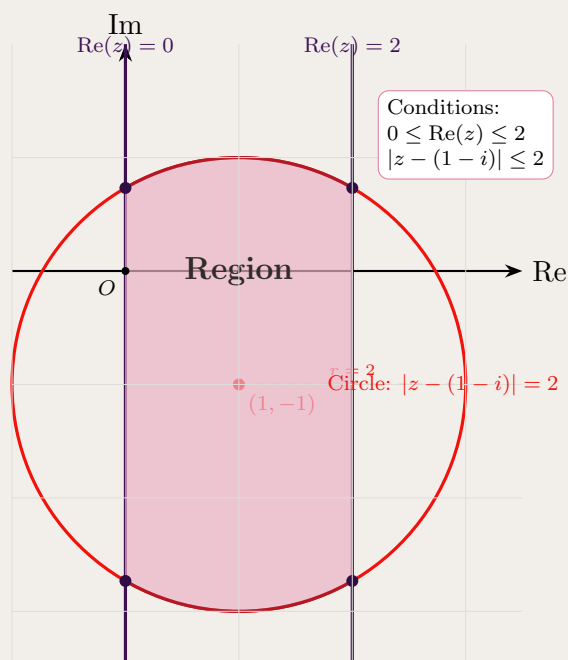
**Problem 3.24: Shading a Region**

On an Argand diagram, shade the region where  $0 \leq \operatorname{Re}(z) \leq 2$  and  $|z - (1 - i)| \leq 2$  both hold.

**Hint:** The first inequality is a vertical strip. The second is a disk centered at  $(1, -1)$  with radius 2.

**Solution 3.24: Sketch**

Draw vertical lines at  $x = 0$  and  $x = 2$ . Draw circle centered at  $(1, -1)$  with radius 2. Shade the intersection.

**Problem 3.25: Shifted Reciprocal Locus**

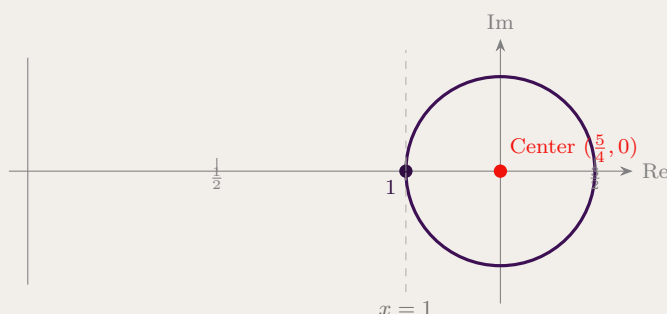
If  $z$  is a complex number, describe the locus defined by  $\Re\left(\frac{1}{z-1}\right) = 2$ .

**Hint:** Write  $z = x + iy$  and set  $w = z - 1 = (x - 1) + iy$ . Compute  $\Re\left(\frac{w}{1}\right)$  by multiplying numerator and denominator by  $\bar{w}$ , then complete the square to reveal the circle; note the locus must pass through  $z = 1$ .

### Solution 3.25: Sketch

$$\Re\left(\frac{1}{z-1}\right) = \frac{x-1}{(x-1)^2 + y^2} = 2 \Rightarrow (x-1)^2 - \frac{1}{2}(x-1) + y^2 = 0$$

Completing the square gives  $(x - \frac{5}{4})^2 + y^2 = (\frac{1}{4})^2$ , a circle centered at  $(\frac{5}{4}, 0)$  with radius  $\frac{1}{4}$  that passes through  $z = 1$  and is tangent to the line  $x = 1$ .



#### Takeaways:

- Translating by 1 then inverting sends vertical lines to circles that pass through the translation point.
- $\Re\left(\frac{1}{z-a}\right) = k \neq 0$  traces a circle through  $z = a$  with center shifted by  $\frac{1}{2k}$  along the real axis from  $a$ .
- Here the circle has center  $(\frac{5}{4}, 0)$ , radius  $\frac{1}{4}$ , and is tangent to the line  $x = 1$ .

### Problem 3.26: Modulus-Argument Division

Given  $\frac{1+i\sqrt{3}}{1+i} = \frac{1+i\sqrt{3}}{2} + \frac{\sqrt{3}-1}{2}i$ , express  $\frac{1+i\sqrt{3}}{1+i}$  in modulus-argument form by converting both numerator and denominator.

**Hint:**  $|1+i\sqrt{3}| = 2$ ,  $\arg(1+i\sqrt{3}) = \pi/3$ ;  $|1+i| = \sqrt{2}$ ,  $\arg(1+i) = \pi/4$ .

### Solution 3.26: Sketch

$$\frac{1+i\sqrt{3}}{1+i} = \frac{2}{\sqrt{2}} \left( \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) \right) = \sqrt{2} \left( \cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right).$$

### Problem 3.27: Power in Polar Form

Let  $\beta = 1 - i\sqrt{3}$ . Express  $\beta^5$  in modulus-argument form.

**Hint:**  $|\beta| = 2$ ,  $\arg(\beta) = -\pi/3$ .

**Solution 3.27: Sketch**

$\beta = 2(\cos(-\pi/3) + i \sin(-\pi/3))$ . Then  $\beta^5 = 32(\cos(-5\pi/3) + i \sin(-5\pi/3)) = 32(\cos(\pi/3) + i \sin(\pi/3))$ .

**Problem 3.28: Deriving Trigonometric Value**

Find the exact value of  $\sin \frac{\pi}{12}$  using the angle subtraction formula. Note that  $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4} = 60^\circ - 45^\circ = 15^\circ$ .

$$\frac{z}{\sqrt{2}} = \frac{1}{2} \cos \frac{\pi}{4}, \quad \frac{z}{\sqrt{2}} = \frac{1}{2} \sin \frac{\pi}{4} \quad \bullet$$

$$\frac{z}{1} = \frac{3}{2} \cos \frac{\pi}{3}, \quad \frac{z}{1} = \frac{3}{2} \sin \frac{\pi}{3} \quad \bullet$$

**Hint:** Use the angle subtraction formula:  $\sin(A - B) = \sin A \cos B - \cos A \sin B$ . Recall the exact values:

**Solution 3.28: Sketch**

We use the fact that  $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$ .

Applying the angle subtraction formula:

$$\begin{aligned} \sin \frac{\pi}{12} &= \sin \left( \frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} \\ &= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \frac{1}{2} \cdot \frac{\sqrt{2}}{2} \\ &= \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} \\ &= \frac{\sqrt{6} - \sqrt{2}}{4} \end{aligned}$$

**Alternative approach using complex numbers:**

Consider the division  $\frac{1+i\sqrt{3}}{1+i}$ . In modulus-argument form:

- Numerator:  $|1 + i\sqrt{3}| = 2$ ,  $\arg(1 + i\sqrt{3}) = \frac{\pi}{3}$
- Denominator:  $|1 + i| = \sqrt{2}$ ,  $\arg(1 + i) = \frac{\pi}{4}$

Therefore:

$$\frac{1 + i\sqrt{3}}{1 + i} = \frac{2}{\sqrt{2}} \left( \cos \left( \frac{\pi}{3} - \frac{\pi}{4} \right) + i \sin \left( \frac{\pi}{3} - \frac{\pi}{4} \right) \right) = \sqrt{2} \left( \cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

Computing the division algebraically:

$$\frac{1 + i\sqrt{3}}{1 + i} = \frac{(1 + i\sqrt{3})(1 - i)}{(1 + i)(1 - i)} = \frac{1 - i + i\sqrt{3} + \sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2} + i \frac{\sqrt{3} - 1}{2}$$

Comparing imaginary parts:  $\sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$ .

**Answer:**  $\sin \frac{\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4}$

**Problem 3.29: Marking Rotated Point**

A point  $P$  represents the complex number  $z = 3 + 2i$  on the Argand diagram. Mark the point  $R$  representing  $R = iz$ .

**Hint:** Multiply by  $i$  rotates  $90^\circ$  counterclockwise. If  $z = a + bi$ , then  $iz = i(a + bi) = (ai + bi^2) = -b + ai$ .

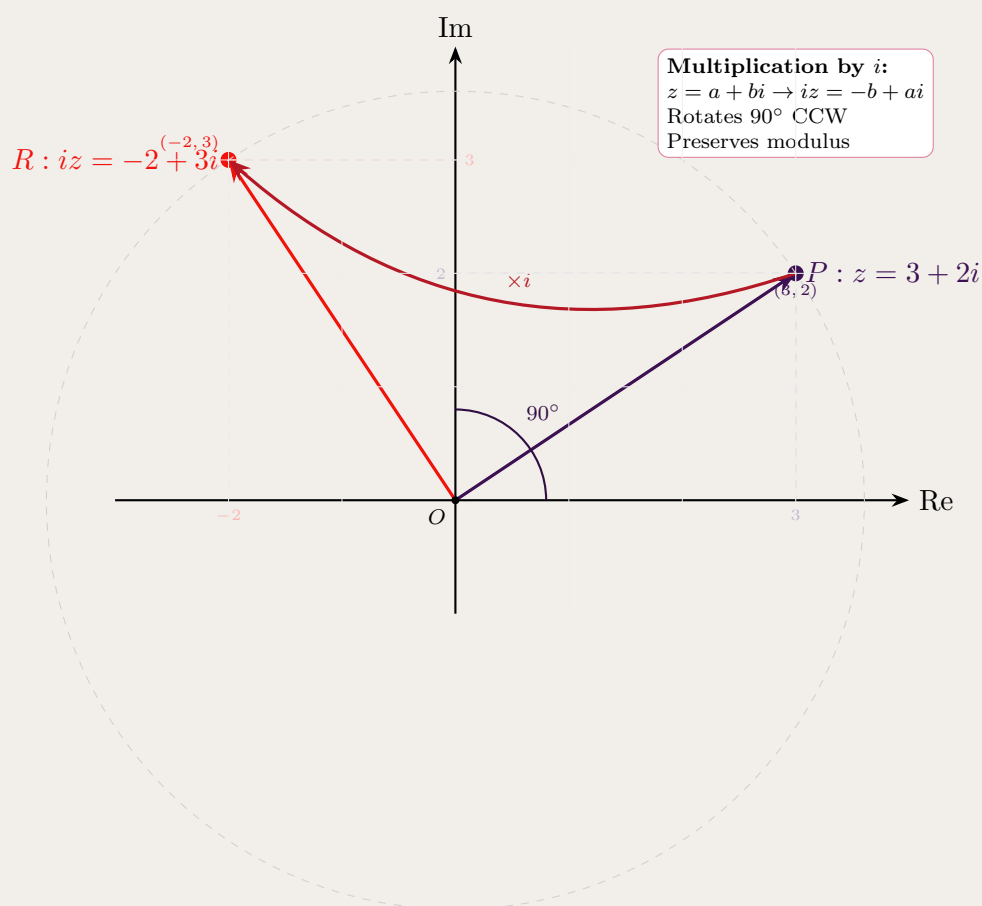
### Solution 3.29: Sketch

If  $z$  is at  $(a, b)$ , then  $iz$  is at  $(-b, a)$ . Draw accordingly.

For  $z = 3 + 2i$  at point  $(3, 2)$ :

$$iz = i(3 + 2i) = 3i + 2i^2 = 3i - 2 = -2 + 3i$$

So  $R$  is at  $(-2, 3)$ .



**Key observation:** Multiplying by  $i$  rotates any complex number by  $90^\circ$  counterclockwise while keeping the same distance from the origin.

### Problem 3.30: Conjugate Root Property

$2 + i$  is a root of  $P(z) = z^3 + rz^2 + sz + 20$  where  $r, s$  are real. State why  $2 - i$  is also a root.

**Hint:** Polynomials with real coefficients have complex roots in conjugate pairs.

**Solution 3.30: Sketch**

Since  $P(z)$  has real coefficients, if  $2 + i$  is a root, then  $\overline{2 + i} = 2 - i$  must also be a root.

**Problem 3.31: Factorizing Over Reals**

Factorise  $P(z) = z^3 + rz^2 + sz + 20$  over the real numbers, given roots  $2 + i$  and  $2 - i$ .

$$z^3 + rz^2 + sz + 20 = (z - (2 + i))(z - (2 - i))(z - t) = (z^2 - 4z + 5)(z - t)$$

**Solution 3.31: Sketch**

$(z - 2 - i)(z - 2 + i) = z^2 - 4z + 5$ . Divide  $P(z)$  by  $z^2 - 4z + 5$  to find the third factor.  $P(z) = (z^2 - 4z + 5)(z + 4)$  after polynomial division.

**Problem 3.32: Vector Addition Point**

Points  $P$  and  $Q$  represent the complex numbers  $z = 2 + i$  and  $w = 1 + 3i$  respectively on the Argand diagram. Mark point  $T$  representing  $T = z + w$ .

**Hint:** Use parallelogram law:  $T$  is the fourth vertex of parallelogram  $OPTQ$  where  $O$  is the origin.

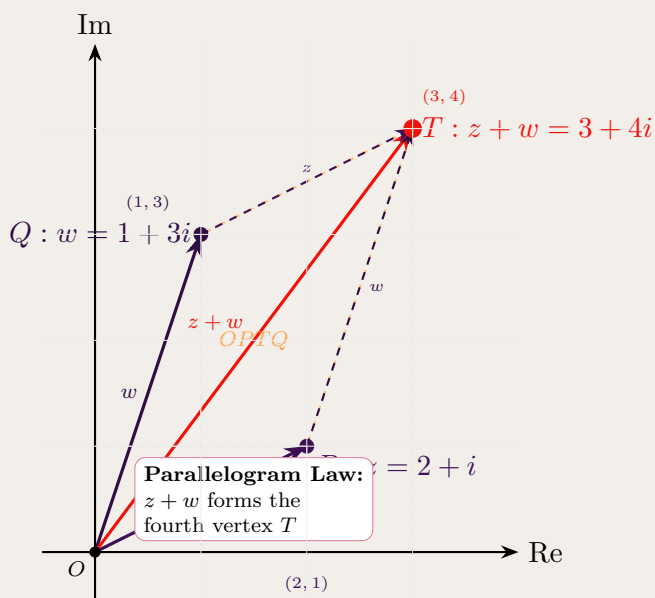
**Solution 3.32: Sketch**

From  $O$  to  $P$  is vector  $z$ , from  $O$  to  $Q$  is vector  $w$ . The sum  $z + w$  is at the opposite corner of the parallelogram.

For  $z = 2 + i$  and  $w = 1 + 3i$ :

$$z + w = (2 + i) + (1 + 3i) = 3 + 4i$$

So  $T$  is at  $(3, 4)$ .

**Key observations:**

- The parallelogram  $OPTQ$  has  $OP$  parallel to  $QT$  (both represent vector  $z$ )
- Similarly,  $OQ$  is parallel to  $PT$  (both represent vector  $w$ )
- Vector addition: starting from  $O$ , go to  $P$  (add  $z$ ), then to  $T$  (add  $w$ )
- Alternatively: starting from  $O$ , go to  $Q$  (add  $w$ ), then to  $T$  (add  $z$ )
- The diagonal  $OT$  represents the sum  $z + w$

**Problem 3.33: Sum in Polar Form**

Let  $z = -2 - 2i$  and  $w = 3 + i$ . Express  $z + w$  in modulus-argument form.

**Hint:** First find  $z + w = 1 - i$ , then convert to polar.

**Solution 3.33: Sketch**

$z + w = 1 - i$ .  $|z + w| = \sqrt{2}$ ,  $\arg(z + w) = -\pi/4$ . So  $z + w = \sqrt{2}(\cos(-\pi/4) + i \sin(-\pi/4))$ .

**Problem 3.34: Division in Mixed Form**

Express  $\frac{z}{w}$  in form  $x + iy$  where  $z = -2 - 2i$ ,  $w = 3 + i$ .

$$\cdot \frac{(i-3)(i+3)}{(3-2i)(3+2i)} = \frac{i+3}{-2-2i} \text{Hint:}$$

**Solution 3.34: Sketch**

$$\frac{(-2-2i)(3-i)}{10} = \frac{-6+2i-6i+2i^2}{10} = \frac{-8-4i}{10} = -\frac{4}{5} - \frac{2}{5}i.$$

**Problem 3.35: Sketching with Argument and Circle**

Sketch the region:  $-\frac{\pi}{4} \leq \arg z \leq 0$  and  $|z - (1 - i)| \leq 1$ .

Hint: The first is a wedge in the fourth quadrant. The second is a disk centered at  $(1, -1)$ .

**Solution 3.35: Sketch**

Draw rays at angles  $-45^\circ$  and  $0^\circ$  from origin. Draw circle centered at  $(1, -1)$  radius 1. Shade intersection.

**Problem 3.36: High Power via De Moivre**

Express  $z^9$  in form  $x + iy$  for  $z = \sqrt{3} - i$ .

$$\text{Hint: } z = 2(\cos(-\pi/6) + i \sin(-\pi/6)).$$

**Solution 3.36: Sketch**

$$z^9 = 2^9(\cos(-3\pi/2) + i \sin(-3\pi/2)) = 512(0 + i) = 512i.$$

**Problem 3.37: Polar Form of Specific Number**

Write  $z = -1 + i\sqrt{3}$  in modulus-argument form.

Hint:  $|z| = 2$ ,  $z$  is in the second quadrant with reference angle  $\pi/3$ .

**Solution 3.37: Sketch**

$$z = 2(\cos(2\pi/3) + i \sin(2\pi/3)).$$

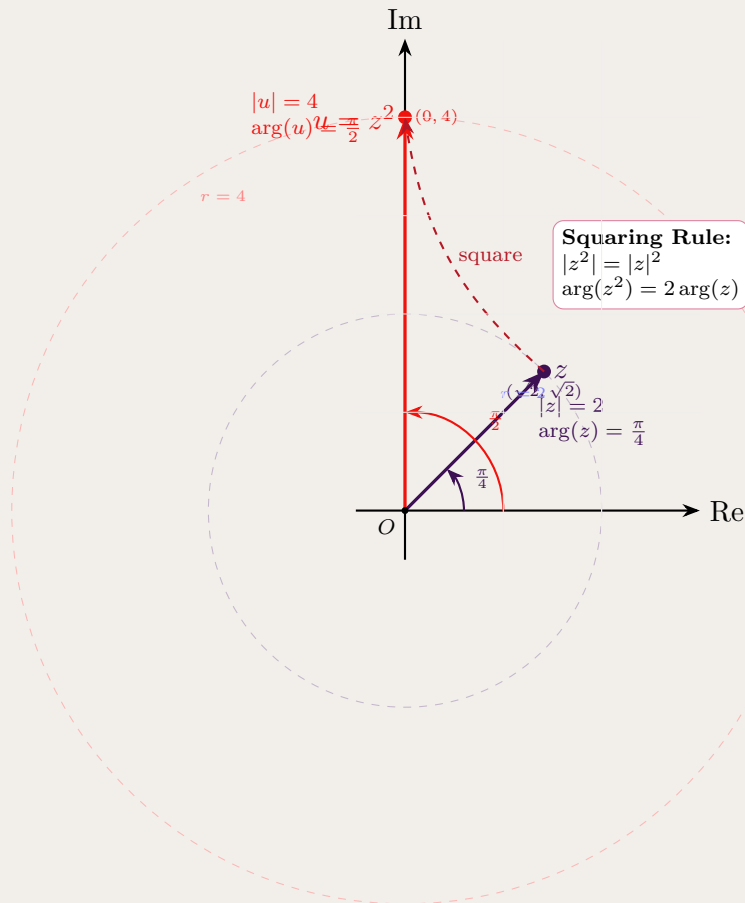
**Problem 3.38: Plotting a Squared Number**

Given  $|z| = 2$ ,  $\arg(z) = \frac{\pi}{4}$ , plot  $u = z^2$ .

**Hint:**  $|u| = |z|^2 = 4, \arg(u) = 2 \arg(z) = \pi/2$ .

### Solution 3.38: Sketch

$u$  has modulus 4 and argument  $\pi/2$ , so  $u = 4i$ . Plot at  $(0, 4)$ .



### Problem 3.39: Fifth Roots of Unity and Cosine Sum

Let  $\omega$  be a non-real root of  $z^5 = 1$

- Show that:  $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$
- Hence, show that:  $\cos\left(\frac{2\pi}{5}\right) + \cos\left(\frac{4\pi}{5}\right) = -\frac{1}{2}$

**Hint:** For part (a): The fifth roots of unity are roots of  $z^5 - 1 = 0$ . Factor as  $(z - 1)(z^4 + z^3 + z^2 + z + 1) = 0$ . For part (b): Use  $\omega = e^{i2\pi/5}$  and  $\omega^4 = e^{i8\pi/5} = e^{-i2\pi/5} = \bar{\omega}$ . Add  $\omega + \omega^4$  using Euler's formula.

**Solution 3.39: Sketch**

(a) Since  $\omega^5 = 1$ , factor  $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1) = 0$ . As  $\omega \neq 1$  (non-real), we have  $z^4 + z^3 + z^2 + z + 1 = 0$ , so  $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ .  $\square$

(b) Let  $\omega = e^{i2\pi/5}$ . Then  $\omega^4 = e^{-i2\pi/5} = \bar{\omega}$  and  $\omega^3 = \bar{\omega}^2$ . From (a):

$$1 + (\omega + \omega^4) + (\omega^2 + \omega^3) = 0 \implies 1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$$

Therefore  $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$ .  $\square$

**Problem 3.40: Cube Roots of Unity Properties**

If  $\omega$  is a complex cube root of unity, show that:

a)  $1 + \omega + \omega^2 = 0$

b)  $1 + \omega + \bar{\omega} = 0$

c)  $(6\omega + 1)(6\omega^2 + 1) = 31$

d)  $(1 + \omega^2)^3(2 + 3\omega + 3\omega^2) = 1$

**Hint:** A complex cube root of unity satisfies  $\omega^3 = 1$  and  $\omega \neq 1$ . Use  $\omega = e^{i2\pi/3}$  and the fundamental identity  $1 + \omega + \omega^2 = 0$  to prove all parts. For (b), note that  $\omega^2 = \bar{\omega}$ . For (c) and (d), expand and repeatedly substitute  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ .

**Solution 3.40: Sketch**

**Key facts:** A complex cube root of unity  $\omega$  satisfies:

- $\omega^3 = 1$
- $\omega = e^{i2\pi/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$
- $\omega^2 = e^{i4\pi/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$
- $\omega^2 = \bar{\omega}$  (conjugates)

**Part (a):** Show  $1 + \omega + \omega^2 = 0$

Since  $\omega$  is a cube root of unity:  $\omega^3 = 1$ , so  $\omega^3 - 1 = 0$ .

Factor:  $(\omega - 1)(\omega^2 + \omega + 1) = 0$

Since  $\omega \neq 1$  (it's a complex root), we must have:

$$\omega^2 + \omega + 1 = 0$$

Therefore:  $1 + \omega + \omega^2 = 0$   $\square$

**Part (b):** Show  $1 + \omega + \bar{\omega} = 0$

Since  $\omega = e^{i2\pi/3}$  and  $\omega^2 = e^{i4\pi/3} = e^{-i2\pi/3}$ , we have:

$$\omega^2 = \bar{\omega}$$

From part (a):  $1 + \omega + \omega^2 = 0$

Substituting  $\omega^2 = \bar{\omega}$ :

$$1 + \omega + \bar{\omega} = 0 \quad \square$$

**Part (c):** Show  $(6\omega + 1)(6\omega^2 + 1) = 31$

Expand the product:

$$\begin{aligned} (6\omega + 1)(6\omega^2 + 1) &= 36\omega \cdot \omega^2 + 6\omega + 6\omega^2 + 1 \\ &= 36\omega^3 + 6\omega + 6\omega^2 + 1 \end{aligned}$$

Use  $\omega^3 = 1$ :

$$= 36(1) + 6\omega + 6\omega^2 + 1 = 36 + 1 + 6(\omega + \omega^2)$$

From part (a):  $\omega + \omega^2 = -1$

Therefore:

$$= 37 + 6(-1) = 37 - 6 = 31 \quad \square$$

**Part (d):** From (a),  $1 + \omega^2 = -\omega$ , so  $(1 + \omega^2)^3 = (-\omega)^3 = -\omega^3 = -1$ . Also,  $2 + 3\omega + 3\omega^2 = 2 + 3(-1) = -1$ . Therefore  $(1 + \omega^2)^3(2 + 3\omega + 3\omega^2) = (-1)(-1) = 1$ .  $\square$

**Problem 3.41: Circle Locus in Complex Plane**

Show that the following relationship geometrically describes a circle in the complex plane:

$$\frac{1}{z} - \frac{1}{\bar{z}} = i$$

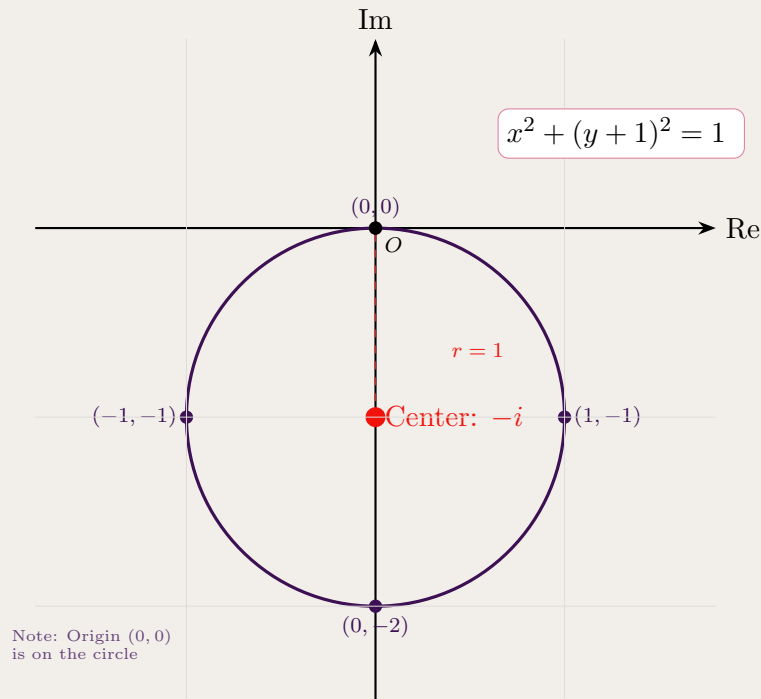
**Hint:** Let  $z = x + iy$  where  $x, y \in \mathbb{R}$ . Express  $\frac{z}{1}$  and  $\frac{\bar{z}}{1}$  in terms of  $x$  and  $y$ , then separate real and imaginary parts. The result should be an equation of the form  $(x - a)^2 + (y - b)^2 = r^2$ .

**Solution 3.41: Sketch**

Let  $z = x + iy$ . Then:

$$\frac{1}{z} - \frac{1}{\bar{z}} = \frac{x - iy}{x^2 + y^2} - \frac{x + iy}{x^2 + y^2} = \frac{-2iy}{x^2 + y^2}$$

Given this equals  $i$ :  $\frac{-2iy}{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 + 2y = 0 \Rightarrow x^2 + (y + 1)^2 = 1$ . This is a circle centered at  $-i$  with radius 1.



**Verification:** The origin  $(0,0)$  satisfies the equation:  $0^2 + (0 + 1)^2 = 1$  ✓

**Geometric interpretation:** For any point  $z$  on this circle, the difference  $\frac{1}{z} - \frac{1}{\bar{z}}$  equals  $i$ . This relates the reciprocals and their conjugates in a specific way that constrains  $z$  to lie on a circle.

**Problem 3.42: Square OABC with Complex Numbers**

$OABC$  is a square with point  $A$  represented by the complex number  $z = 2 + i$ .

- Let  $w$  represent the complex number at point  $C$ . Prove that:  $z^2 + w^2 = 0$
- Find the complex numbers represented by  $B$  and  $C$ .
- Find the complex number represented by the vector  $\overrightarrow{AC}$

**Hint:** For a square  $OABC$  with  $O$  at the origin,  $C$  is obtained by rotating  $A$  by  $90^\circ$  counterclockwise, so  $w = iz$ . For part (a), substitute and use  $i^2 = -1$ . For part (b),  $B$  is the fourth vertex, obtained by  $B = A + C$  (vector addition).

**Solution 3.42: Sketch**

**Given:** Square  $OABC$  with  $O$  at origin and  $A$  at  $z = 2 + i$ .

**Part (a):** Prove  $z^2 + w^2 = 0$

For a square with one vertex at the origin  $O$  and adjacent vertex at  $A$  (represented by  $z$ ), the next vertex  $C$  (going counterclockwise) is obtained by rotating  $OA$  by  $90^\circ$  counterclockwise.

Rotation by  $90^\circ$  is achieved by multiplying by  $i$ :

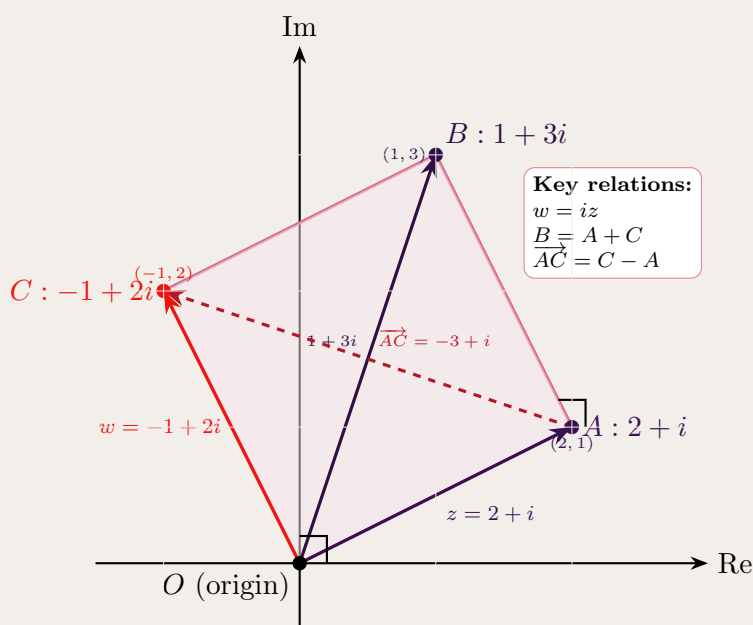
$$w = iz$$

Now calculate  $z^2 + w^2$ :

$$\begin{aligned} z^2 + w^2 &= z^2 + (iz)^2 \\ &= z^2 + i^2 z^2 \\ &= z^2 + (-1)z^2 \\ &= z^2 - z^2 \\ &= 0 \quad \square \end{aligned}$$

(b)  $C$ :  $w = iz = i(2 + i) = 2i - 1 = -1 + 2i$ .  $B$ :  $B = A + C = (2 + i) + (-1 + 2i) = 1 + 3i$ .

(c)  $\overrightarrow{AC} = C - A = (-1 + 2i) - (2 + i) = -3 + i$ .

**Problem 3.43: Square ABCD Vertices with Complex Numbers**

$ABCD$  is a square in the complex plane. The vertices  $A$  and  $B$  represent the complex numbers  $9 + i$  and  $4 + 13i$  respectively. Find the complex numbers corresponding to vertices  $C$  and  $D$ .

**Hint:** The vector from  $A$  to  $B$  is  $\overrightarrow{AB} = B - A = (4 + 13i) - (9 + i) = -5 + 12i$ . To find  $C$ , rotate  $\overrightarrow{AB}$  by  $90^\circ$  counterclockwise (multiply by  $i$ ) and add to  $B$ . Similarly for  $D$ .

**Solution 3.43: Sketch**

$\overrightarrow{AB} = B - A = (4 + 13i) - (9 + i) = -5 + 12i$ . Rotate by  $90^\circ$ :  $\overrightarrow{BC} = i(-5 + 12i) = -12 - 5i$ . Thus  $C = B + \overrightarrow{BC} = (4 + 13i) + (-12 - 5i) = -8 + 8i$ . Similarly,  $D = A + \overrightarrow{BC} = (9 + i) + (-12 - 5i) = -3 - 4i$ .

**Answers:**  $C = -8 + 8i$ ,  $D = -3 - 4i$ .

**Problem 3.44: Square ABCD Vector Relationship**

The points  $A, B, C$  and  $D$  on the Argand diagram represent the complex numbers  $a, b, c$  and  $d$  respectively. The points form a square  $ABCD$ .

By using vectors, or otherwise, show that  $c = (1 + i)d - ia$ .

**Hint:** Use the fact that in a square, consecutive sides are equal in length and perpendicular. The key relationship is  $\overrightarrow{DC} = i \cdot \overrightarrow{AD}$  (rotation by  $90^\circ$ ).

**Solution 3.44: Proof**

In square  $ABCD$ ,  $\overrightarrow{DC} = i \cdot \overrightarrow{AD}$  (rotation by  $90^\circ$ ). Thus:

$$c - d = i(d - a) = id - ia \implies c = d + id - ia = (1 + i)d - ia \quad \square$$

**3.3 Advanced Complex Numbers Problems****Problem 3.45: Complex Region with Exclusion**

Sketch the region on the Argand diagram where  $|z - \bar{z}| < 2$  and  $|z - 1| \geq 1$  hold simultaneously.

**Hint:**  $|z - \bar{z}| = 2|y|$ , so the first inequality gives  $|y| < 1$ . The second is the exterior of a circle.

**Solution 3.45: Sketch**

The region  $|y| < 1$  is a horizontal strip  $-1 < y < 1$ . The region  $|z - 1| \geq 1$  is outside the circle centered at  $(1, 0)$  with radius 1. Shade the intersection: horizontal strip with circular hole.

**Problem 3.46: Isosceles Right Triangle**

Triangle  $ABC$  is represented by  $z_1, z_2, z_3$ . It is isosceles and right-angled at  $B$ . Explain why  $(z_1 - z_2)^2 = -(z_3 - z_2)^2$ .

**Hint:** The vectors  $BA$  and  $BC$  are perpendicular and equal in length. Multiplication by  $i$  rotates by  $90^\circ$ .

**Solution 3.46: Sketch**

$BA = z_1 - z_2$ ,  $BC = z_3 - z_2$ . Since  $BA \perp BC$  and  $|BA| = |BC|$ , we have  $BA = \pm i \cdot BC$ . Squaring:  $(z_1 - z_2)^2 = (\pm i)^2 (z_3 - z_2)^2 = -1 \cdot (z_3 - z_2)^2$ .

**Problem 3.47: Square from Triangle**

Three vertices of a square  $ABCD$  in the complex plane are represented by complex numbers  $z_1$  (point  $A$ ),  $z_2$  (point  $B$ ), and  $z_3$  (point  $C$ ). Given that consecutive sides of a square are perpendicular and equal in length, find the complex number  $z_4$  representing vertex  $D$  in terms of  $z_1, z_2, z_3$ .

**Hint:** Since  $ABCD$  is a square, side  $AB$  is perpendicular to side  $BC$ , and  $|AB| = |BC|$ . This means  $BA = i \cdot BC$  (or  $-i \cdot BC$ ). Use the same relationship for sides  $CD$  and  $DA$ .

**Solution 3.47: Sketch**

For a square  $ABCD$ , consecutive sides are equal in length and perpendicular.

The vector from  $B$  to  $A$  is  $z_1 - z_2$ , and from  $B$  to  $C$  is  $z_3 - z_2$ .

Since  $BA \perp BC$  and  $|BA| = |BC|$ , we have:

$$z_1 - z_2 = i(z_3 - z_2) \quad (\text{rotating } BC \text{ by } 90^\circ \text{ counterclockwise})$$

Similarly, for the square to close, the vector from  $C$  to  $D$  must equal the vector from  $B$  to  $C$  rotated  $90^\circ$ :

$$z_4 - z_3 = i(z_2 - z_3)$$

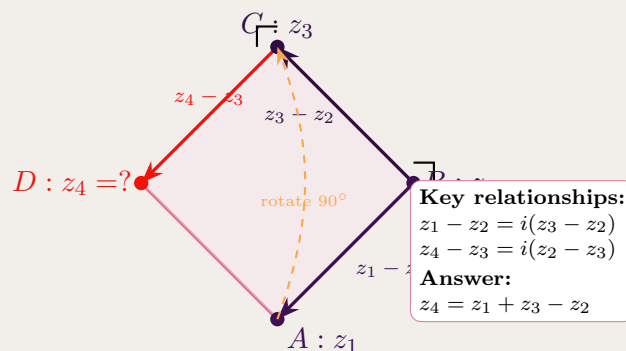
Solving for  $z_4$ :

$$\begin{aligned} z_4 &= z_3 + i(z_2 - z_3) \\ &= z_3 + iz_2 - iz_3 \\ &= z_2 \cdot i + z_3(1 - i) \\ &= iz_2 + z_3(1 - i) \end{aligned}$$

**Alternative formula:** Using the parallelogram property, we can also write:

$$z_4 = z_1 + z_3 - z_2$$

This works because  $\vec{AD} = \vec{BC}$ , so  $z_4 - z_1 = z_3 - z_2$ .



**Verification:** Both formulas are equivalent:

$$\begin{aligned} iz_2 + z_3(1 - i) &= iz_2 + z_3 - iz_3 \\ &= z_3 + i(z_2 - z_3) \end{aligned}$$

And using  $z_1 - z_2 = i(z_3 - z_2)$ , we get  $z_1 = z_2 + i(z_3 - z_2) = iz_3 + z_2(1 - i)$ .

Therefore:  $z_4 = z_1 + z_3 - z_2$

**Problem 3.48: Factoring Polynomial**

Express  $P(x) = x^4 - 12x^3 + 59x^2 - 138x + 130$  as product of quadratic factors, given roots  $3 + i$  and  $3 + 2i$ .

**Hint:** The roots are  $3 \pm i$  and  $3 \pm 2i$ . Group conjugate pairs.

### Solution 3.48: Sketch

$(x - (3 + i))(x - (3 - i)) = (x - 3)^2 + 1 = x^2 - 6x + 10$ .  $(x - (3 + 2i))(x - (3 - 2i)) = (x - 3)^2 + 4 = x^2 - 6x + 13$ .  
Thus  $P(x) = (x^2 - 6x + 10)(x^2 - 6x + 13)$ .

### Problem 3.49: Real Root Count

Deduce that  $x^4 - 3x^3 + 5x^2 + 7x - 8 = 0$  has exactly two real roots.

**Hint:** A degree-4 polynomial with real coefficients has either 0, 2, or 4 real roots. Check behavior or use Descartes' rule.

### Solution 3.49: Sketch

Complex roots come in conjugate pairs. If there are 4 complex roots, they form 2 pairs, leaving 0 real roots. If 1 pair of complex roots, then 2 real roots. Since the polynomial has sign changes, it must have at least one positive real root. Testing values or using intermediate value theorem confirms exactly 2 real roots.

### Problem 3.50: Vector Addition and Angles

Consider a sequence of unit complex numbers defined by:

$$z_n = \cos(\alpha + n\beta) + i \sin(\alpha + n\beta) = e^{i(\alpha + n\beta)}$$

for  $n = 0, 1, 2, 3, \dots$

On the Argand diagram, we construct points by cumulative vector addition:

$$\begin{aligned} P_0 &= z_0 \\ P_1 &= z_0 + z_1 \\ P_2 &= z_0 + z_1 + z_2 \\ P_3 &= z_0 + z_1 + z_2 + z_3 \\ &\vdots \end{aligned}$$

These points form a polygonal path. Using vector addition, prove that all external angles of this polygon are equal to  $\beta$ . That is, show that  $\theta_0 = \theta_1 = \theta_2 = \dots = \beta$ , where  $\theta_n$  is the external angle at vertex  $P_n$ .

**Hint:** Each  $z_n$  is a unit vector at angle  $\alpha + n\beta$ . The angle between consecutive vectors is constant. The external angle is the amount the direction turns from one segment to the next.

**Solution 3.50: Sketch****Understanding the geometry:**

The polygonal path is formed by placing unit vectors  $z_0, z_1, z_2, \dots$  tip-to-tail. Each vector  $z_n$  is a unit complex number (lies on the unit circle) with argument  $\alpha + n\beta$ .

**Step 1: Arguments of consecutive vectors**

For any consecutive pair of vectors:

$$\begin{aligned}\arg(z_n) &= \alpha + n\beta \\ \arg(z_{n+1}) &= \alpha + (n+1)\beta = \alpha + n\beta + \beta\end{aligned}$$

The difference in arguments is:

$$\arg(z_{n+1}) - \arg(z_n) = \beta$$

**Step 2: Geometric interpretation**

The segment from  $P_n$  to  $P_{n+1}$  is represented by the vector  $z_{n+1}$  (pointing in direction  $\arg(z_{n+1})$ ).

The segment from  $P_{n-1}$  to  $P_n$  is represented by the vector  $z_n$  (pointing in direction  $\arg(z_n)$ ).

The external angle  $\theta_n$  at vertex  $P_n$  is the angle by which the path turns. This is precisely the difference in the directions of consecutive vectors:

$$\theta_n = \arg(z_{n+1}) - \arg(z_n) = \beta$$

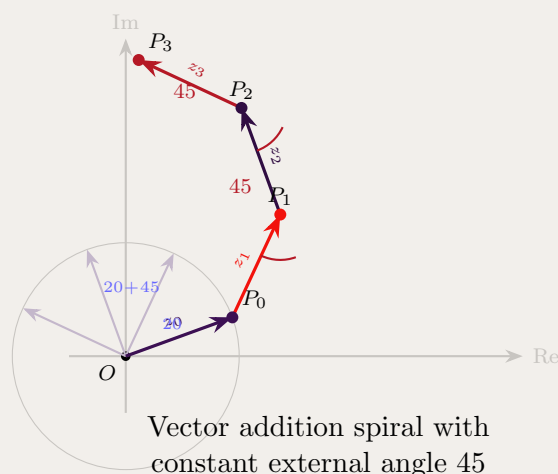
**Step 3: Conclusion**

Since this difference is constant for all  $n$ , we have:

$$\theta_0 = \theta_1 = \theta_2 = \dots = \beta$$

All external angles equal  $\beta$ , which means the polygon turns by the same angle at each vertex.

**Special case:** If  $\beta = \frac{2\pi}{n}$ , the path closes after  $n$  steps, forming a regular  $n$ -sided polygon.



The diagram shows unit vectors  $z_0, z_1, z_2, z_3$  (shown faintly from origin) being added tip-to-tail to form the polygonal path  $P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3$ . Each turn angle (external angle) is  $\beta = 45^\circ$ .

**Problem 3.51: Modulus via Trigonometry**

Show  $|z| = 2 \sin \theta$  for  $z = 1 - \cos 2\theta + i \sin 2\theta$ .

**Hint:** Use  $1 - \cos 2\theta = 2 \sin^2 \theta$  and  $\sin 2\theta = 2 \sin \theta \cos \theta$ .

**Solution 3.51: Sketch**

$|z|^2 = (1 - \cos 2\theta)^2 + \sin^2 2\theta = (2 \sin^2 \theta)^2 + (2 \sin \theta \cos \theta)^2 = 4 \sin^4 \theta + 4 \sin^2 \theta \cos^2 \theta = 4 \sin^2 \theta (\sin^2 \theta + \cos^2 \theta) = 4 \sin^2 \theta$ . Thus  $|z| = 2 |\sin \theta|$ .

**Problem 3.52: Euler's Formula Identity**

Show  $e^{in\theta} + e^{-in\theta} = 2 \cos(n\theta)$ .

**Hint:** Use  $e^{i\phi} = \cos \phi + i \sin \phi$ .

**Solution 3.52: Sketch**

$e^{in\theta} = \cos(n\theta) + i \sin(n\theta)$  and  $e^{-in\theta} = \cos(n\theta) - i \sin(n\theta)$ . Adding:  $e^{in\theta} + e^{-in\theta} = 2 \cos(n\theta)$ .

**Problem 3.53: Modulus and Angle Calculation with Complex Exponentials**

Let  $z$  be the complex number  $z = e^{\frac{i\pi}{6}}$  and  $w$  be the complex number  $w = e^{\frac{3i\pi}{4}}$ .

(i) By first writing  $z$  and  $w$  in Cartesian form, or otherwise, show that

$$|z + w|^2 = \frac{4 - \sqrt{6} + \sqrt{2}}{2}.$$

(ii) The complex numbers  $z$ ,  $w$  and  $z + w$  are represented in the complex plane by the vectors  $\vec{OA}$ ,  $\vec{OB}$  and  $\vec{OC}$  respectively, where  $O$  is the origin.

Show that  $\angle AOC = \frac{7\pi}{24}$ .

(iii) Deduce that  $\cos \frac{7\pi}{24} = \frac{\sqrt{8 - 2\sqrt{6} + 2\sqrt{2}}}{4}$ .

**Hint:** For (i): Convert to Cartesian form and compute  $|z + w|^2 = x^2 + y^2$ . For (ii): Since  $|z| = |w| = 1$ , the parallelogram  $OACB$  is a rhombus, and the diagonal bisects the angle. For (iii): Use the relationship  $|z + w| = 2 \cos(\angle AOC)$  in a rhombus.

**Solution 3.53: Full Solution****Part (i)**

First, we convert  $z$  and  $w$  to Cartesian form:

$$z = e^{\frac{i\pi}{6}} = \cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$w = e^{\frac{3i\pi}{4}} = \cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

Now, find  $z + w$ :

$$z + w = \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}\right) + i\left(\frac{1}{2} + \frac{\sqrt{2}}{2}\right) = \frac{\sqrt{3} - \sqrt{2}}{2} + i\frac{1 + \sqrt{2}}{2}$$

Calculate the square of the modulus  $|z + w|^2 = x^2 + y^2$ :

$$\begin{aligned} |z + w|^2 &= \left(\frac{\sqrt{3} - \sqrt{2}}{2}\right)^2 + \left(\frac{1 + \sqrt{2}}{2}\right)^2 \\ &= \frac{3 - 2\sqrt{6} + 2}{4} + \frac{1 + 2\sqrt{2} + 2}{4} \\ &= \frac{5 - 2\sqrt{6} + 3 + 2\sqrt{2}}{4} \\ &= \frac{8 - 2\sqrt{6} + 2\sqrt{2}}{4} \\ &= \frac{4 - \sqrt{6} + \sqrt{2}}{2} \quad \checkmark \end{aligned}$$

**Part (ii)**

Note that  $|z| = |e^{\frac{i\pi}{6}}| = 1$  and  $|w| = |e^{\frac{3i\pi}{4}}| = 1$ . Since the moduli are equal ( $OA = OB = 1$ ), the parallelogram  $OACB$  formed by vector addition is a **rhombus**.

In a rhombus, the main diagonal  $\overrightarrow{OC}$  bisects the angle between the adjacent sides  $\overrightarrow{OA}$  and  $\overrightarrow{OB}$ .

Find  $\angle AOB$ :

$$\begin{aligned} \arg(z) &= \frac{\pi}{6} = \frac{4\pi}{24}, \quad \arg(w) = \frac{3\pi}{4} = \frac{18\pi}{24} \\ \angle AOB &= \arg(w) - \arg(z) = \frac{18\pi}{24} - \frac{4\pi}{24} = \frac{14\pi}{24} \end{aligned}$$

Since  $\overrightarrow{OC}$  bisects  $\angle AOB$ , the angle between  $\overrightarrow{OA}$  and  $\overrightarrow{OC}$  is half of  $\angle AOB$ :

$$\angle AOC = \frac{1}{2} \times \frac{14\pi}{24} = \frac{7\pi}{24} \quad \checkmark$$

**Part (iii)**

We use the geometry of the rhombus. The length of the diagonal  $OC$  can be calculated using the property that in a rhombus with side length 1 and angle  $\theta$  between the diagonal and the side:

$$OC = 2 \times OA \times \cos(\angle AOC)$$

Substituting the known values:

$$|z + w| = 2(1) \cos\left(\frac{7\pi}{24}\right) = 2 \cos\left(\frac{7\pi}{24}\right)$$

Squaring both sides:

$$|z + w|^2 = 4 \cos^2\left(\frac{7\pi}{24}\right)$$

Substitute the result from Part (i):

$$\begin{aligned} \frac{4 - \sqrt{6} + \sqrt{2}}{2} &= 4 \cos^2\left(\frac{7\pi}{24}\right) \\ \cos^2\left(\frac{7\pi}{24}\right) &= \frac{4 - \sqrt{6} + \sqrt{2}}{8} = \frac{8 - 2\sqrt{6} + 2\sqrt{2}}{16} \end{aligned}$$

Taking the square root (noting the angle is acute, so cosine is positive):

$$\cos\left(\frac{7\pi}{24}\right) = \sqrt{\frac{8 - 2\sqrt{6} + 2\sqrt{2}}{16}} = \frac{\sqrt{8 - 2\sqrt{6} + 2\sqrt{2}}}{4} \quad \checkmark$$

### Problem 3.54: Equilateral Triangle

Let  $z_1$  be a non-zero complex number represented by point  $A$  on the Argand diagram, and let  $z_2 = e^{i\pi/3}z_1$  be represented by point  $B$ . The origin is represented by point  $O$ . Prove that triangle  $\triangle OAB$  is equilateral.

**Hint:** Multiplication by  $e^{i\pi/3}$  rotates  $z_1$  by angle  $\pi/3 = 60^\circ$  while preserving its modulus. Therefore  $|z_1| = |z_2|$  and the angle  $\angle AOB = \pi/3$ .

### Solution 3.54: Sketch

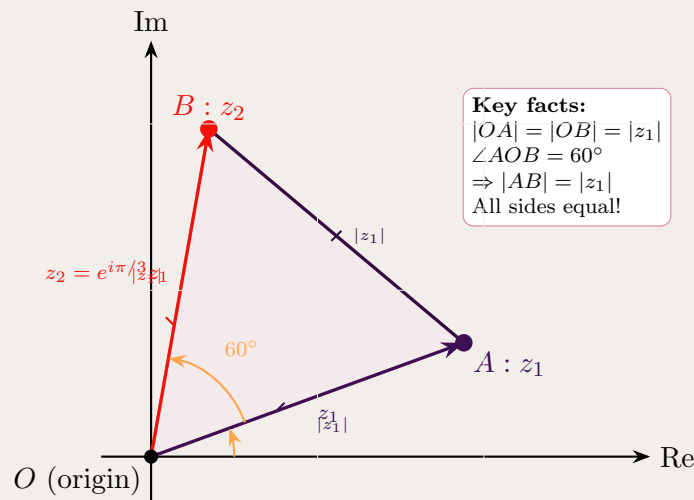
Given  $z_2 = e^{i\pi/3}z_1$ , we show all three sides of triangle  $OAB$  are equal.

**Step 1:** Since  $|e^{i\pi/3}| = 1$ , we have  $|OA| = |z_1|$  and  $|OB| = |z_2| = |e^{i\pi/3}z_1| = |z_1|$ .

**Step 2:** Since  $z_2 = e^{i\pi/3}z_1$ , the angle  $\angle AOB = \arg(z_2) - \arg(z_1) = \frac{\pi}{3} = 60^\circ$ .

**Step 3:** By law of cosines:  $|AB|^2 = |z_1|^2 + |z_1|^2 - 2|z_1|^2 \cos(\pi/3) = 2|z_1|^2 - |z_1|^2 = |z_1|^2$ , so  $|AB| = |z_1|$ .

**Conclusion:**  $|OA| = |OB| = |AB| = |z_1|$ , so triangle  $OAB$  is equilateral.



**Alternative:**  $|AB| = |z_2 - z_1| = |z_1||e^{i\pi/3} - 1| = |z_1| \cdot |(1/2 + i\sqrt{3}/2) - 1| = |z_1|\sqrt{1/4 + 3/4} = |z_1|$ .  $\square$

### Problem 3.55: Algebraic Identity

Consider an equilateral triangle  $\triangle OAB$  where  $O$  is the origin,  $A$  represents complex number  $z_1$ , and  $B$  represents  $z_2 = e^{i\pi/3}z_1$  (i.e.,  $z_2$  is obtained by rotating  $z_1$  by  $60^\circ$ ). Prove the algebraic identity:

$$z_1^2 + z_2^2 = z_1 z_2$$

**Hint:** Let  $\omega = e^{i\pi/3}$ , so  $z_2 = \omega z_1$ . Show that  $\omega$  satisfies  $\omega^2 - \omega + 1 = 0$  by using the fact that  $\omega^3 = e^{i\pi} = -1$ .

**Solution 3.55: Sketch**

Let  $\omega = e^{i\pi/3}$  (a primitive 6th root of unity). Then  $z_2 = \omega z_1$ .

**Left-hand side:**

$$\begin{aligned} z_1^2 + z_2^2 &= z_1^2 + (\omega z_1)^2 \\ &= z_1^2 + \omega^2 z_1^2 \\ &= z_1^2(1 + \omega^2) \end{aligned}$$

**Right-hand side:**

$$\begin{aligned} z_1 z_2 &= z_1 \cdot \omega z_1 \\ &= \omega z_1^2 \end{aligned}$$

**To prove:** We need to show that  $1 + \omega^2 = \omega$ , or equivalently:

$$\omega^2 - \omega + 1 = 0$$

**Key observation:** Since  $\omega = e^{i\pi/3}$ , we have:

$$\omega^3 = e^{i\pi/3 \cdot 3} = e^{i\pi} = -1$$

Therefore:

$$\omega^3 + 1 = 0$$

This factors as:

$$(\omega + 1)(\omega^2 - \omega + 1) = 0$$

Since  $\omega = e^{i\pi/3} = \cos 60^\circ + i \sin 60^\circ = \frac{1}{2} + i \frac{\sqrt{3}}{2} \neq -1$ , we must have:

$$\omega^2 - \omega + 1 = 0$$

Therefore  $1 + \omega^2 = \omega$ , which gives us:

$$z_1^2(1 + \omega^2) = z_1^2 \cdot \omega = \omega z_1^2 = z_1 z_2$$

Thus:  $z_1^2 + z_2^2 = z_1 z_2$   $\square$

**Alternative direct verification:** We can also verify by direct calculation:

$$\begin{aligned} \omega^2 &= e^{i2\pi/3} = \cos 120^\circ + i \sin 120^\circ = -\frac{1}{2} + i \frac{\sqrt{3}}{2} \\ 1 + \omega^2 &= 1 + \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2}\right) = \frac{1}{2} + i \frac{\sqrt{3}}{2} = \omega \quad \checkmark \end{aligned}$$

**Problem 3.56: Finding Non-Real Roots**

Find the two complex roots of  $P(x) = x^5 - 10x^2 + 15x - 6$ .

**Hint:** Test for rational roots first (like  $x = 1$ ). Factor out linear factors, then solve the remaining polynomial.

**Solution 3.56: Sketch**

$P(1) = 1 - 10 + 15 - 6 = 0$ , so  $(x - 1)$  is a factor. Polynomial division gives  $P(x) = (x - 1)(x^4 + x^3 + x^2 - 9x + 6)$ . Continue factoring or use numerical/algebraic methods to find complex roots.

**Problem 3.57: Expansion Result**

Using De Moivre's theorem and the binomial expansion, derive the formula for  $\cos 5\theta$  in terms of  $\cos \theta$ . Hence show that:

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

**Hint:** Use Euler's formula:  $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$ . Expand the left side using the binomial theorem, then equate real parts. Express  $\sin^2 \theta$  in terms of  $\cos \theta$  using  $\sin^2 \theta = 1 - \cos^2 \theta$ .

**Solution 3.57: Sketch**

By De Moivre's theorem:

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

Expand the left side using binomial theorem:

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= \sum_{k=0}^5 \binom{5}{k} \cos^{5-k} \theta (i \sin \theta)^k \\ &= \binom{5}{0} \cos^5 \theta + \binom{5}{1} \cos^4 \theta (i \sin \theta) + \binom{5}{2} \cos^3 \theta (i \sin \theta)^2 \\ &\quad + \binom{5}{3} \cos^2 \theta (i \sin \theta)^3 + \binom{5}{4} \cos \theta (i \sin \theta)^4 + \binom{5}{5} (i \sin \theta)^5 \\ &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ &\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \end{aligned}$$

Separate real and imaginary parts:

Real part:

$$\text{Re} = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

This equals  $\cos 5\theta$ .

Express in terms of  $\cos \theta$  only:

Use  $\sin^2 \theta = 1 - \cos^2 \theta$ :

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= (1 + 10 + 5) \cos^5 \theta + (-10 - 10) \cos^3 \theta + 5 \cos \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

Therefore:  $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$   $\square$

**Problem 3.58: Symmetry Identity**

Show  $\alpha^k + \alpha^{-k} = 2 \cos k\theta$  where  $\alpha = \cos \theta + i \sin \theta$ .

**Hint:**  $e^{i\theta} = \cos \theta + i \sin \theta$  and  $e^{-i\theta} = \cos \theta - i \sin \theta$ .

**Solution 3.58: Sketch**

$\alpha^k = e^{ik\theta} = \cos k\theta + i \sin k\theta$ ,  $\alpha^{-k} = e^{-ik\theta} = \cos k\theta - i \sin k\theta$ . Adding:  $\alpha^k + \alpha^{-k} = 2 \cos k\theta$ .

**Problem 3.59: Equilateral Triangle Centroid**

Let  $\triangle ABC$  be an equilateral triangle with vertices  $A, B, C$  represented by complex numbers  $a, b, c$  respectively, oriented anticlockwise.

Let  $\omega = e^{i2\pi/3} = \cos(2\pi/3) + i\sin(2\pi/3)$

be a primitive cube root of unity. Prove that the centroid of the triangle is at the origin if and only if:

$$a + b\omega + c\omega^2 = 0$$

**Hint:** For an equilateral triangle centered at the origin, the vertices are related by  $120^\circ$  rotations. Use the fact that  $\omega = e^{i2\pi/3}$  satisfies  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ .

**Solution 3.59: Sketch**

**Key facts about  $\omega = e^{i2\pi/3}$ :**

- $\omega^3 = e^{i2\pi} = 1$  (cube root of unity)
- $1 + \omega + \omega^2 = 0$  (sum of cube roots of unity)
- $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

**If the triangle is centered at origin:**

For an equilateral triangle centered at the origin with one vertex at  $a$ , the other two vertices are obtained by rotating by  $120^\circ$  and  $240^\circ$ :

$$b = \omega a \quad \text{and} \quad c = \omega^2 a$$

Now compute:

$$\begin{aligned} a + b\omega + c\omega^2 &= a + (\omega a) \cdot \omega + (\omega^2 a) \cdot \omega^2 \\ &= a + \omega^2 a + \omega^4 a \\ &= a(1 + \omega^2 + \omega^4) \end{aligned}$$

Since  $\omega^3 = 1$ , we have  $\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$ .

Therefore:

$$a + b\omega + c\omega^2 = a(1 + \omega^2 + \omega) = a(1 + \omega + \omega^2)$$

Using the fundamental identity  $1 + \omega + \omega^2 = 0$ :

$$a + b\omega + c\omega^2 = a \cdot 0 = 0$$

**Conversely:** If  $a + b\omega + c\omega^2 = 0$  and the triangle is equilateral, then its centroid is at the origin. The centroid is at  $\frac{a+b+c}{3}$ . For an equilateral triangle with the given property, we can show:

$$a + b + c = 0$$

This follows because if  $a + b\omega + c\omega^2 = 0$ , and we also have the relations for an equilateral triangle, then the centroid must be at the origin.

Therefore:  $a + b\omega + c\omega^2 = 0$   $\square$

**Geometric interpretation:** This identity expresses the fact that in an equilateral triangle centered at the origin, the weighted sum of the vertices (with weights  $1, \omega, \omega^2$ ) is zero, reflecting the rotational symmetry of the configuration.

**Problem 3.60: Real Sum and Product**

Given that  $z + w$  and  $zw$  are both real, prove that either  $z = \bar{w}$  or  $\text{Im}(z) = \text{Im}(w) = 0$ .

**Hint:** Let  $z = a + bi$  and  $w = c + di$ . Use the conditions that  $z + w$  and  $zw$  are real to derive equations for  $b$  and  $d$  in terms of  $a$  and  $c$ .

**Solution 3.60: Proof**

Let  $z = a + bi$ ,  $w = c + di$  where  $a, b, c, d \in \mathbb{R}$ .

**Step 1:** If  $z + w$  is real, then  $\text{Im}(z + w) = b + d = 0$ , so  $d = -b$ . (Eq 1)

**Step 2:** If  $zw$  is real, then  $\text{Im}(zw) = \text{Im}[(a + bi)(c + di)] = ad + bc = 0$ . (Eq 2)

**Step 3:** Substitute Eq 1 into Eq 2:  $a(-b) + bc = 0 \implies b(c - a) = 0$ .

**Case 1:** If  $b = 0$ , then  $d = -b = 0$ , so  $\text{Im}(z) = \text{Im}(w) = 0$  (both real).

**Case 2:** If  $b \neq 0$ , then  $c = a$ . With  $d = -b$ :  $w = a - bi = \bar{z}$ , so  $z = \bar{w}$ .

**Conclusion:** Either  $\text{Im}(z) = \text{Im}(w) = 0$  (both real) or  $z = \bar{w}$  (conjugates).  $\square$

**Problem 3.61: Complex Argument Region**

The complex numbers  $w$  and  $z$  both have modulus 1, and  $\frac{\pi}{2} < \text{Arg}\left(\frac{z}{w}\right) < \pi$ , where  $\text{Arg}$  denotes the principal argument.

For real numbers  $x$  and  $y$ , consider the complex number  $\frac{xz + yw}{z}$ .

On an  $xy$ -plane, clearly sketch the region that contains all points  $(x, y)$  for which

$$\frac{\pi}{2} < \text{Arg}\left(\frac{xz + yw}{z}\right) < \pi.$$

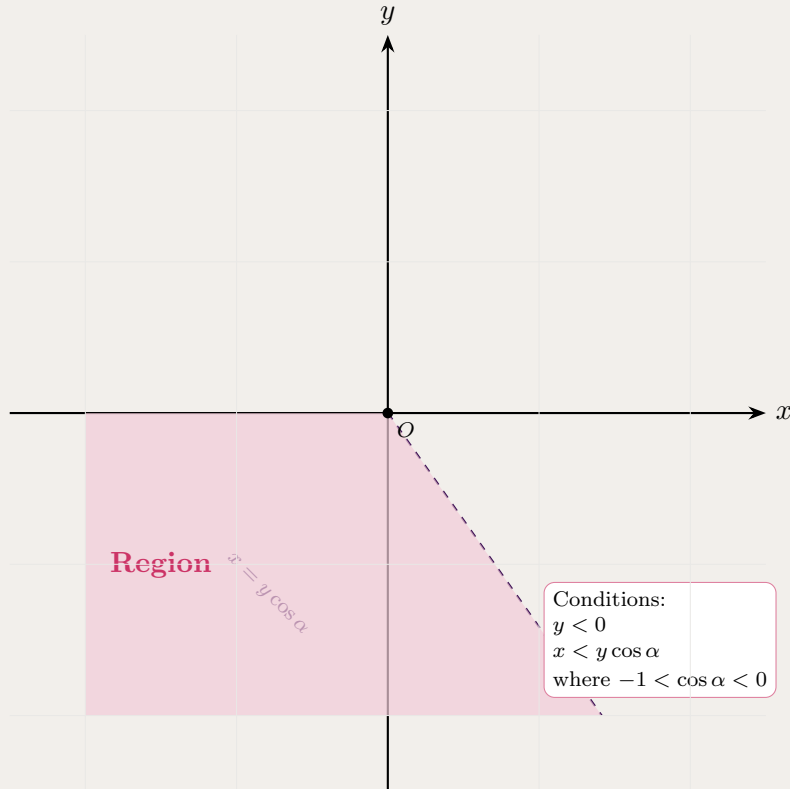
**Hint:** Simplify  $\frac{xz + yw}{z} = x + y\frac{w}{z}$ . Let  $\alpha = \text{Arg}(w/z)$  where  $\frac{\pi}{2} < \alpha < \pi$ . Then  $w/z = e^{i\alpha}$ . The expression becomes  $x + ye^{i\alpha}$ . For the argument to be in the given range, the resulting complex number must lie in the second quadrant.

**Solution 3.61: Sketch**

Simplify:  $\frac{xz+yw}{z} = x + y\frac{w}{z}$ . Since  $\frac{\pi}{2} < \text{Arg}(z/w) < \pi$ , we have  $-\pi < \text{Arg}(w/z) < -\frac{\pi}{2}$ , so  $w/z = e^{i\alpha}$  where  $-\pi < \alpha < -\frac{\pi}{2}$ . Thus:

$$x + y(\cos \alpha + i \sin \alpha) = (x + y \cos \alpha) + i(y \sin \alpha)$$

For this in Q2 (second quadrant):  $x + y \cos \alpha < 0$  and  $y \sin \alpha > 0$ . Since  $\sin \alpha < 0$  and  $\cos \alpha < 0$  in Q3, we need  $y < 0$  and  $x < y \cos \alpha$ . The region is a wedge below the  $x$ -axis bounded by the line  $x = y \cos \alpha$ .

**Problem 3.62: Geometric Series with  $n$ -th Roots**

If  $z = \cos \frac{\pi}{n} + i \sin \frac{\pi}{n}$ , and  $n$  is a positive integer, prove:

- (i)  $1 + z + z^2 + \dots + z^{2n-1} = 0$
- (ii)  $1 + z + z^2 + \dots + z^{n-1} = 1 + i \cot \left( \frac{\pi}{2n} \right)$

**Hint:** For part (i): Note that  $z = e^{i\pi/n}$ , so  $z_{2n} = e^{i2\pi} = 1$ . Use the geometric series formula. For part (ii): Split the sum at  $z_{n-1}$  and use the fact that  $z_n = e^{i\pi} = -1$ .

**Solution 3.62: Proof**

**Setup:** Given  $z = \cos \frac{\pi}{n} + i \sin \frac{\pi}{n} = e^{i\pi/n}$

**Part (i):** Prove  $1 + z + z^2 + \dots + z^{2n-1} = 0$

Calculate  $z^{2n}$ :

$$z^{2n} = \left(e^{i\pi/n}\right)^{2n} = e^{i \cdot 2\pi} = 1$$

Therefore  $z$  is a  $2n$ -th root of unity.

The sum  $S = 1 + z + z^2 + \dots + z^{2n-1}$  is a geometric series with first term 1, common ratio  $z$ , and  $2n$  terms.

Using the geometric series formula:

$$S = \frac{z^{2n} - 1}{z - 1} = \frac{1 - 1}{z - 1} = \frac{0}{z - 1} = 0$$

(Note: Since  $z = e^{i\pi/n} \neq 1$  for  $n \geq 2$ , the denominator is non-zero.)

Therefore:  $1 + z + z^2 + \dots + z^{2n-1} = 0$   $\square$

**Part (ii):** Prove  $1 + z + z^2 + \dots + z^{n-1} = 1 + i \cot\left(\frac{\pi}{2n}\right)$

Let  $S_n = 1 + z + z^2 + \dots + z^{n-1}$

From part (i), we know:

$$1 + z + z^2 + \dots + z^{2n-1} = 0$$

We can split this sum:

$$(1 + z + \dots + z^{n-1}) + (z^n + z^{n+1} + \dots + z^{2n-1}) = 0$$

The second sum can be factored:

$$z^n + z^{n+1} + \dots + z^{2n-1} = z^n(1 + z + \dots + z^{n-1}) = z^n \cdot S_n$$

Since  $z^n = e^{i\pi} = -1$ :

$$S_n + (-1) \cdot S_n = 0$$

$$S_n - S_n = 0$$

Using geometric series:  $S_n = \frac{z^n - 1}{z - 1} = \frac{-2}{z - 1}$  (since  $z^n = -1$ ). With  $z - 1 = (\cos \frac{\pi}{n} - 1) + i \sin \frac{\pi}{n}$ , multiply by conjugate. Denominator:  $2(1 - \cos \frac{\pi}{n}) = 4 \sin^2 \frac{\pi}{2n}$ . Using  $\cos \frac{\pi}{n} - 1 = -2 \sin^2 \frac{\pi}{2n}$  and  $\sin \frac{\pi}{n} = 2 \sin \frac{\pi}{2n} \cos \frac{\pi}{2n}$ , we get:

$$S_n = \frac{4 \sin^2 \frac{\pi}{2n} + i \cdot 4 \sin \frac{\pi}{2n} \cos \frac{\pi}{2n}}{4 \sin^2 \frac{\pi}{2n}} = 1 + i \cot \frac{\pi}{2n} \quad \square$$

Therefore:  $1 + z + z^2 + \dots + z^{n-1} = 1 + i \cot\left(\frac{\pi}{2n}\right)$   $\square$

**Problem 3.63: Fifth Roots of  $-1$  and Cosine Values**

By finding the fifth roots of  $-1$ , find the exact values of  $\cos \frac{\pi}{5}$  and  $\cos \frac{3\pi}{5}$ .

**Hint:** The fifth roots of  $-1$  are solutions to  $z^5 = -1$ , or equivalently  $z^5 + 1 = 0$ . Use  $(1 + z)(1 + z^2 + z^4) = 0$ . The polynomial  $z^5 + 1$  factors as  $(1 + z)(1 + z^2 + z^4)$ , so  $z = e^{i\pi/5}$  for  $k = 0, 1, 2, 3, 4$ . The polynomial  $z^5 + 1$  factors as  $(1 + z)(1 + z^2 + z^4)$ , so  $z = e^{i\pi/5}$  for  $k = 0, 1, 2, 3, 4$ . Use the quartic to find relationships involving  $\cos \frac{\pi}{5}$  and  $\cos \frac{3\pi}{5}$ .

**Solution 3.63: Sketch**

**Step 1:** Solve  $z^5 = -1 = e^{i\pi}$  using De Moivre:  $z = e^{i\pi(2k+1)/5}$  for  $k = 0, 1, 2, 3, 4$ .

Five roots:  $z_0 = e^{i\pi/5}$ ,  $z_1 = e^{i3\pi/5}$ ,  $z_2 = -1$ ,  $z_3 = e^{-i3\pi/5} = \bar{z}_1$ ,  $z_4 = e^{-i\pi/5} = \bar{z}_0$ .

**Step 2:** Factor:  $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1) = 0$ . The four non-real roots  $z_0, z_1, \bar{z}_0, \bar{z}_1$  satisfy  $z^4 - z^3 + z^2 - z + 1 = 0$ .

**Step 3:** Use Vieta's formulas

Sum of roots:  $(z_0 + \bar{z}_0) + (z_1 + \bar{z}_1) = 1 \implies 2\cos\frac{\pi}{5} + 2\cos\frac{3\pi}{5} = 1 \implies \cos\frac{\pi}{5} + \cos\frac{3\pi}{5} = \frac{1}{2}$ .

Sum of products (coeff. of  $z^2$ ): Using  $z_0 z_4 = |z_0|^2 = 1$ ,  $z_1 z_3 = |z_1|^2 = 1$ , and simplifying other products gives:  $\alpha\beta = -\frac{1}{4}$  where  $\alpha = \cos\frac{\pi}{5}$  and  $\beta = \cos\frac{3\pi}{5}$ .

From  $\alpha + \beta = \frac{1}{2}$  and  $\alpha\beta = -\frac{1}{4}$ , solve  $4t^2 - 2t - 1 = 0$  to get  $t = \frac{1 \pm \sqrt{5}}{4}$ . Since  $\cos\frac{\pi}{5} > 0$  and  $\cos\frac{3\pi}{5} < 0$ :

$$\boxed{\cos\frac{\pi}{5} = \frac{1 + \sqrt{5}}{4}} \quad \text{and} \quad \boxed{\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}}$$

**Problem 3.64: Euler's Formula and Integration**

(i) Show that for any integer  $n$ ,  $e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$ .

(ii) By expanding  $(e^{i\theta} + e^{-i\theta})^4$ , show that

$$\cos^4\theta = \frac{1}{8}(\cos(4\theta) + 4\cos(2\theta) + 3).$$

(iii) Hence, or otherwise, find  $\int_0^{\pi/2} \cos^4\theta \, d\theta$ .

**Hint:** For part (i): Use Euler's formula  $e^{i\phi} = \cos\phi + i\sin\phi$ . For part (ii): Use part (i) to write  $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ , then raise to the 4th power using the binomial theorem. For part (iii): Integrate the expression from part (ii) term by term.

**Solution 3.64: Sketch**

**Part (i):** Show  $e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$

Using Euler's formula:

$$\begin{aligned} e^{in\theta} &= \cos(n\theta) + i\sin(n\theta) \\ e^{-in\theta} &= \cos(-n\theta) + i\sin(-n\theta) = \cos(n\theta) - i\sin(n\theta) \end{aligned}$$

Adding:

$$e^{in\theta} + e^{-in\theta} = 2\cos(n\theta) \quad \square$$

**(ii)** From (i),  $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ , so  $\cos^4\theta = \frac{(e^{i\theta} + e^{-i\theta})^4}{16}$ . Expand using binomial:  $(e^{i\theta} + e^{-i\theta})^4 = e^{i4\theta} + 4e^{i2\theta} + 6 + 4e^{-i2\theta} + e^{-i4\theta} = 2\cos 4\theta + 8\cos 2\theta + 6$ . Thus  $\cos^4\theta = \frac{1}{8}(\cos 4\theta + 4\cos 2\theta + 3)$ .  $\square$

$$\text{(iii)} \quad \int_0^{\pi/2} \cos^4\theta \, d\theta = \frac{1}{8} \left[ \frac{\sin 4\theta}{4} + 2\sin 2\theta + 3\theta \right]_0^{\pi/2} = \frac{1}{8} \cdot \frac{3\pi}{2} = \boxed{\frac{3\pi}{16}}$$

## 4 Conclusion

Complex numbers are a core component of the HSC Mathematics Extension 2 course. Mastery comes from repeated, reflective practice across all forms—Cartesian, polar, and exponential—and understanding their geometric interpretations. Use these problems to sharpen the ability to convert between forms, apply De Moivre's theorem, interpret loci geometrically, and communi-

cate complete mathematical reasoning. Best of luck with your studies and HSC examinations!

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